Four-Color Boundary-Order Notes

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Abstract

This note records the current status of the boundary-order layer in the Lean formalization of the Goertzel four-color proof route. The key point is now formalized as an explicit regression: the present `PlaneEmbeddingWithBoundary` API only exposes unordered face-boundary incidence, and this is too weak to recover the ordered face-boundary geometry used downstream by the selected-boundary and annulus arguments. Moreover, even the current cyclic edge-list and cyclic edge/vertex-walk surrogates are still strictly weaker than an honest actual facial closed-walk source interface. The newest obstruction goes in the opposite direction: once the honest closed-walk source is used to extract a separated two-component annulus boundary split, the current all-roots-on-the-outer-boundary peel package cannot use that same split. A later diamond-with-triangle forcing-edge calibration now also shows that repaired previous-boundary witness geometry does not imply the interior-dual root-distance package.

1 Current source interfaces

The development now distinguishes ten source-facing or immediately derived boundary-order layers above `PlaneEmbeddingWithBoundary`:

1. `OrderedFaceBoundaryData`: each ambient face boundary is listed as a linear chain of distinct edges with shared endpoints between consecutive edges.
2. `OrderedFaceBoundaryVertexData`: the same linear edge order together with one explicit chosen shared primal endpoint for each adjacent pair of boundary edges.
3. `OrderedFaceBoundaryVertexSequenceData`: the same linear edge order together with one explicit ordered list of shared primal vertices, aligned with the boundary edge list itself.
4. `CyclicOrderedFaceBoundaryData`: the same linear data, plus a wrap condition saying the last and first boundary edges also share an endpoint.
5. `CyclicOrderedFaceBoundaryVertexData`: the cyclic edge order together with one explicit chosen shared primal endpoint for each adjacent pair in the closed cycle.
6. `CyclicOrderedFaceBoundaryVertexSequenceData`: the same cyclic edge order together with one explicit cyclic list of shared primal vertices, again stored in boundary order.
7. `FaceBoundaryClosedVertexWalkGeometry`: a bundled cyclic alternating edge/vertex walk, where each boundary edge is incident both to its own listed vertex and to the next listed vertex in cyclic order.
8. `FaceBoundaryClosedRunGeometry`: the same cyclic information repackaged as one explicit closed boundary run per ambient face, i.e. a pure boundary-walk layer.

9. `FaceBoundaryClosedWalkGeometry`: a stronger bundled source object recording for each face an actual nonempty closed trail in the graph whose traversed edges are exactly the face boundary.

10. FourColor specializations of these structures, together with selected boundary arc data.

These interfaces are intentional temporary surrogates for honest embedding-side face-boundary walk order. They isolate exactly what the current weak API fails to provide.

2 Negative examples

Three explicit regressions are now in Lean.

First, the disconnected two-edge example shows that unordered face-boundary incidence alone does not determine any endpoint-sharing order at all. This lives in `Mettapedia/GraphTheory/OrderedPlanarEmbeddingWithBoundaryRegression.lean` and is reused in `Mettapedia/GraphTheory/FourColor/PlanarBoundaryFaceBoundaryRunRegression.lean`.

Second, the three-edge path example shows that linear ordered data is strictly weaker than cyclic ordered data and than the equivalent pure closed-run layer. The unique face boundary can be listed as `[p01, p12, p23]` or its reverse, so linear ordered data exists. But the path endpoints do not share a primal endpoint, so no cyclic closure is possible. This proves that the cyclic source layer is a genuine strengthening rather than just a repackaging.

Third, the three-edge star example shows that the current cyclic source layers are still weaker than an honest facial closed-walk source. The unique face boundary can be listed cyclically as the three spokes, because every consecutive pair shares the hub vertex. So cyclic ordered data exists, and hence also the equivalent bundled cyclic closed-run and cyclic edge/vertex-walk surrogates. But the star graph is acyclic, so it admits no nonempty actual closed trail at all. This proves that the new `FaceBoundaryClosedWalkGeometry` target is a genuine strengthening over the current cyclic surrogates on the weak embedding API.

On the FourColor side, the same star example now also lifts to the cyclic selected-boundary shells `PlanarBoundaryCyclicOrderedFaceArcEmbeddingData` and `PlanarBoundarySelectedBoundaryArcGeometry`, because in the one-face embedding every boundary edge is selected. So even these route-facing cyclic selected-boundary interfaces are still weaker than the honest actual facial closed-walk source. In particular, no generic theorem of the form "current cyclic selected-boundary shell implies honest facial closed-walk geometry" can hold over the present weak embedding API.

3 Architectural consequence

The current proof-route architecture is now sharper:

- unordered face-boundary incidence is too weak;
- linear ordered face-boundary data is enough for the present run-geometry and bundled selected-arc shells;
- the new proof-relevant linear and cyclic boundary-step vertex layers are equivalent theorem surfaces over the current ordered and cyclic edge-order layers, but they remember the actual shared primal vertices between successive boundary edges;

- the new ordered and cyclic boundary-step vertex sequence layers go one step further by storing those chosen shared vertices in the same order as the boundary walk itself, making them a closer source proxy to an honest embedding side facial walk than the pair-indexed chooser interface;
- the new bundled cyclic edge/vertex walk layer is equivalent to the cyclic vertex-sequence interface, but packages the same information in the explicit alternating walk form that future honest embedding-side derivations are most likely to target directly;
- the new honest face-boundary closed-walk layer is strictly stronger than the cyclic ordered / cyclic closed-run / cyclic edge/vertex-walk surrogate layers over the current weak API;
- cyclic ordered data is a closer proxy to genuine planar face-boundary order and is strictly stronger;
- the new pure closed-run layer gives an equivalent "boundary walk" presentation of the cyclic source data, while the new edge/vertex walk layer refines that presentation by keeping the shared primal vertices explicitly in cyclic order;
- the annulus boundary-connectivity and construction shells now also accept the split same-embedding interface "cyclic alternating edge/vertex face-boundary walk + selected-boundary arc on each induced linear run" directly, so downstream theorem endpoints no longer have to manually unwrap the cyclic walk layer before using the current selected-arc reductions;
- the stronger honest source surface "facial closed walk + selected-boundary arc on each induced linear run" now also lowers directly to the current cyclic ordered face-arc shell and to the same annulus boundary-connectivity / boundary-support / positive-frontier / construction endpoints, so future work can target actual facial walks without first repackaging them by hand as cyclic surrogate data;
- the honest facial closed-walk lowering now constructs the cyclic edge/vertex boundary-walk layer directly from the actual walk darts: the boundary edges are the dart edges in order, and the shared vertices are the dart terminal vertices. This removes an avoidable detour through the cyclic edge-order surrogate and makes the proof-relevant vertex data traceable to the source walk itself;
- the current same-embedding annulus theorem surfaces "ordered face-arc data", "cyclic ordered face-arc data", "cyclic edge/vertex walk + selected-boundary arc", and "honest facial closed walk + selected-boundary arc", each combined with the live interior-dual adjacency-distance peel data and the current positive-frontier hypotheses, now also lower directly to the pure annulus decomposition endpoint, removing one more manual construction-to-decomposition composition step from the theorem-4.9 route;
- on the construction side, the same four honest boundary-order source surfaces now also lower directly to the weaker "FacePartitionData" shell, replacing the stronger per-face non-peeled selected-boundary-contact hypothesis by the more honest finite-set face-cover obligation;
- on the live boundary-root adjacency-distance source route, that finite face-cover obligation is now discharged internally: the interior-dual root package already proves that every non-peeled face is a boundary-root face and hence meets selected boundary support. The generic

construction bridge now lowers boundary-component reachability plus this interior-dual package directly to the boundary-support, face-partition, and positive-frontier shells, leaving the positive current-frontier facts as the explicit remaining frontier-side inputs;

- the positive-frontier package now has a checked finite-descent consequence: whenever one annulus stage is strictly later than another, the surviving current-face set has strictly smaller cardinality. Adjacent stages therefore form a genuine finite decreasing chain, so the current-frontier assumptions are not merely interface markers but yield a measurable progress invariant. Since the final current-face set is nonempty and stage zero covers all ambient faces under the face-partition hypothesis, the package also proves the global finite bound that the number of collars is at most the number of ambient faces;

- the shifted boundary-root adjacency-distance construction now also has an exact obstruction theorem for parent-witness orientation: failure of strict parent orientation is equivalent to the existence of a peeled face with shifted annulus distance zero. This records precisely why the first collar must be excluded before the parent-oriented well-founded witness surface can be used;

- the same boundary-root adjacency-distance source now also has a checked constructive repair: a root-distance annulus construction that keeps the unshifted interior-dual distance. On peeled faces, the shifted collar-indexing distance plus one is exactly this root-distance height, and the root-distance construction reaches the parent-oriented well-founded witness surface without any first-collar exclusion. Thus the formal route separates the collar-indexing construction from the parent-orientation construction instead of trying to make one shifted distance serve both purposes;

- on the outer-boundary-root side, the formal interface now also proves that every chosen root face already lies in the ambient outer-boundary face set. So the unresolved face-cover obstruction is genuinely about non-root faces, not about localizing the root set itself;

- the honest closed-walk source now also exposes a structural obstruction to the current all-outer-root route: if `PlanarBoundaryOuterBoundaryRootAdjDistancePeelData` uses the same two-component boundary split extracted from `PlanarBoundaryClosedWalkAnnulusBoundarySource`, then it is inconsistent. The theorem `false_of_closedWalkAnnulusBoundarySource_and_outerBoundaryRootAdjDistancePeelData_sameBoundaryData` proves the fixed-embedding contradiction, and `not_exists_embedding_closedWalkAnnulusBoundarySource_and_outerBoundaryRootAdjDistancePeelData_sameBoundaryData` packages it at graph level. This means the surviving root-distance route needs a two-sided/root-per-boundary formulation or a different boundary split before it can be honestly sourced from closed facial walks;

- the two-sided replacement formulation is now a Lean interface, not just an instruction for future work. The file `Theorem49PlanarBoundaryAnnulusRootInteriorDual.lean` introduces `PlanarBoundaryAnnulusRootAdjDistancePeelData`, whose roots may touch either extracted boundary component. The constructor from `PlanarBoundaryAnnulusBoundaryData` plus generic `InteriorDualBoundaryRootAdjDistancePeelData` and the raw cover/separated-root constructor over `outerAmbientBoundary` union `innerAmbientBoundary` both lower to the existing boundary-root pipeline. The same file also has direct closed-walk constructors from

`PlanarBoundaryClosedWalkAnnulusBoundarySource` plus generic interior-dual data into the two-sided target, so this is the current positive target for closed-walk sourcing;

- the theorem-4.9 synthesis layer now factors through that same generic interior-dual route. The file `Theorem49Synthesis.lean` defines the core `Theorem49BoundaryRootSynthesis` package directly from generic `PlanarBoundaryInteriorDualBoundaryRootAdjDistancePeelData`, and then re-exports it as both `Theorem49OuterBoundaryRootSynthesis` and `Theorem49AnnulusRootSynthesis`. So once boundary-order work supplies the generic interior-dual route, the purified theorem-4.9 endpoint is already available on the two-sided annulus-root surface without any extra synthesis repackaging;

- the route file `PlanarBoundaryClosedWalkSourceRoute.lean` now closes that positive interface explicitly for the honest source objects already used in this note: theorems `exists_theorem49AnnulusRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_direct` and `exists_theorem49AnnulusRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_interiorDualBoundaryRootAdjDistancePeelData_direct` show that explicit closed-walk or successor-cycle boundary-order data plus generic interior-dual data already reaches the two-sided theorem-4.9 synthesis endpoint directly. The same route file now also exposes the more direct certificate-side corollaries `admitsPlanarBoundaryAttachedRootedFacePeelCertificate_of_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_direct`, `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_direct`, and the corresponding successor-cycle versions, so once the generic boundary-root interior-dual package is in hand the honest source shells can land directly on the attached-certificate and boundary-root synthesis surfaces without going back through annulus-construction parent orientation;

- the annulus-root surface now also proves the missing two-sided geometric forcing fact: any ambient face meeting either distinguished annulus boundary component is automatically non-peeled and hence a root face for the generic interior-dual adjacency-distance package. In particular, `exists_root_mem_outerBoundaryFaces_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData` and `exists_root_mem_innerBoundaryFaces_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData` show that honest closed-walk annulus source data plus generic interior-dual data already forces one chosen root face on each extracted annulus boundary component;

- the same source file now also exposes the upstream parent-orientation surface explicitly. The same-embedding theorem `parentWitnessOrientation_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_rootDistance` and the graph-level theorems `admitsPlanarBoundaryAnnulusConstructionParentWitnessOrientation_of_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_direct` and `admitsPlanarBoundaryAnnulusConstructionParentWitnessOrientation_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_interiorDualBoundaryRootAdjDistancePeelData_direct` show that honest closed-walk and successor-cycle boundary-order data already yield `PlanarBoundaryAnnulusConstruction.ParentWitnessOrientation` through the root-distance annulus construction, before any annihilator or synthesis packaging is applied;

- the synthesis layer is now also exposed one step earlier, at the raw annulus-split interface itself: `exists_theorem49AnnulusRootSynthesis_of_exists_boundaryData_and_interiorDualBoundaryRootAdjDistancePeelData` and its nonempty counterpart show that once a single embedding already carries `PlanarBoundaryAnnulusBoundaryData` and generic interior-dual rooted peel data, the theorem-4.9 endpoint is available without first repackaging the same data through closed-walk source fields;

- the concrete annulus-construction route now also reaches the algebraic theorem-4.9 bridge directly. The new file `Theorem49PlanarBoundaryAnnulusConstructionSpanning.lean` proves `boundaryZeroAnnihilatorTrivialForEmbedding_of_planarBoundaryAnnulusConstruction_of_parentWitness`: construction data plus the explicit `ParentWitnessOrientation` hypothesis already implies the fixed-embedding annihilator endpoint "boundary-zero plus Definition 4.8 orthogonality implies zero". From that generic bridge hypothesis it derives the corrected span identity on the projected generator subspace, the corresponding theorem-4.9 target equality and inclusion, pointwise separation on the purified endpoint carrier, kernel-triviality of the pairing map, the resulting finrank inequality, and a raw Kempe-span preimage theorem. So once boundary-order work delivers honest parent-witness orientation on the annulus construction, the route can already enter the algebraic theorem-4.9 bridge without rebuilding the older interior-dual boundary-root package;

- the same algebraic bridge has now been pushed down to the boundary-aware witness-cover surfaces themselves. `Theorem49SpanningGap.lean` proves `boundaryZeroAnnihilatorTrivialForEmbedding_of_planarBoundaryHeightOrderedFacePeelWitnessData` and `boundaryZeroAnnihilatorTrivialForEmbedding_of_planarBoundaryCollarLayerFacePeelWitnessData` `Theorem49SpanningGap.lean` also exposes graph-level height-ordered and collar-layer wrappers for the ordinary and relative submodule-duality interfaces; the chain-dot versions build the duality package directly from the same-embedding target/source subspaces and their identification lemmas. `Theorem49KempeGeneratorSpan.lean` fixes the canonical Definition 4.8 generator side for both surfaces. Its graph-level wrappers `exists_theorem49_kempeGeneratorSubspace_spanning_of_admitsPlanarBoundaryHeightOrderedFacePeelWitnessData` and `exists_theorem49_kempeGeneratorSubspace_spanning_of_admitsPlanarBoundaryCollarLayerFacePeelWitnessData` instantiate the concrete submodule-duality package using the actual Kempe-closure generator subspace, removing the separate orthogonality-to-annihilator obligation from this canonical source path. Then `Theorem49BoundaryProjection.lean` proves the corresponding projected-source equalities `projectedKempeClosureGeneratorSubspace_eq_planarBoundaryZeroSubmodule_of_planarBoundaryHeightOrderedFacePeelWitnessData` and `projectedKempeClosureGeneratorSubspace_eq_planarBoundaryZeroSubmodule_of_planarBoundaryCollarLayerFacePeelWitnessData`, with matching boundary-zero Kirchhoff target equality and inclusion theorems. Thus height-ordered or finite collar-layer witness-cover data is enough to enter the corrected projected-generator theorem-4.9 source statement directly. The same projection file now also provides finite projected-generator representations, raw Definition 4.8 generator preimages before boundary erasure, and graph-level wrappers from `AdmitsPlanarBoundaryHeightOrderedFacePeelWitnessData` and `AdmitsPlanarBoundaryCollarLayerFacePeelWitnessData`. Those wrappers now cover the explicit-family target identity, projected and explicit-family inclusion forms, raw Definition 4.8 generator-span preimages, and finite raw-generator projection witnesses for every boundary-zero Kirchhoff target chain. They now also include the chain-dot separation theorem

$$W_0(H) \cap (\text{projectedKempeClosureGeneratorSubspace})^{\perp} = 0,$$

plus detector and pointwise orthogonality iff forms, for height-ordered and finite collar-layer data. Thus the downstream route can use those witness surfaces without rebuilding the interior-dual rooted package or assuming the extra raw-source quotient disjointness hypotheses. The weakest explicit local-remainder collar surface now has the same kind of direct annihilator access: `Theorem49PlanarBoundaryCollarPeeling.lean` proves `zero_on_allEdges_of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData_of_annihilatesKempeClosureGeneratorFamily` and `zero_on_allEdges_of_exists_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData_of_annihilatesKempeClosureGeneratorFamily`. So local-remainder data can consume Definition 4.8 generator-family orthogonality directly, before being forgotten to ordinary collar-layer data. `Theorem49SpanningGap.lean` and `Theorem49BoundaryProjection.lean` now keep that local-remainder surface visible through the algebraic projection step. In particular `span_projectedKempeClosureGeneratorFamily_eq_planarBoundaryZeroSubmodule_of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData` identifies the projected Definition 4.8 span with the selected-boundary-zero subspace, and `exists_projectedKempeClosureGeneratorFamily_finset_sum_of_mem_theorem49BoundaryZeroKirchhoffSubspace_of_admitsPlanarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData` packages the graph-level finite-combination theorem for the concrete boundary-zero Kirchhoff target. The same file now records the raw Definition 4.8 source projection too: `exists_kempeClosureGeneratorSubspace_preimage_boundaryZeroProjection_of_mem_theorem49BoundaryZeroKirchhoffSubspace_of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData` gives a raw generator-subspace preimage, while `exists_kempeClosureGeneratorFamily_finset_sum_boundaryZeroProjection_of_mem_theorem49BoundaryZeroKirchhoffSubspace_of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData` gives a finite unprojected generator-family sum whose boundary-erased projection is the target chain. It now also has the matching Kirchhoff-retaining raw source theorem `exists_kempeClosureGeneratorFamily_finset_sum_boundaryZeroProjection_and_mem_kirchhoffSubmodule_of_mem_theorem49BoundaryZeroKirchhoffSubspace_of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData`: under the selected-boundary avoidance hypothesis, the same kind of raw finite Definition 4.8 sum can be chosen inside the Kirchhoff submodule at the chosen target vertices. The graph-level admissibility theorem exposes this endpoint without converting the local-remainder witness into interior-dual root-distance data. The local-remainder source has also been pushed to the raw quotient/error form. `exists_kempeClosureGeneratorFamily_finset_sum_boundaryZeroProjection_eq_boundaryZeroProjection_of_mem_kirchhoffSubmodule_of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData` represents every raw Kirchhoff chain, after selected-boundary erasure, by a finite raw

Definition 4.8 generator-family sum that is itself Kirchhoff. The boundary-supported and kernel-decomposition forms then isolate the remaining error as a Kirchhoff chain supported on the selected boundary, equivalently in the kernel of the boundary-erasure map restricted to the Kirchhoff submodule. The same raw representative plus quotient/error package is now available at the positive projected synthesis endpoint itself via `Theorem49BoundaryRootNonemptyProjectedSynthesis.rawKirchhoffRepresentative_and_boundaryKernelDecomposition`. This theorem simply exposes the two relevant fields already inside the nested `Theorem49BoundaryRootSynthesis`, so downstream boundary-order source arguments can consume them through the same-embedding projected endpoint rather than importing another raw-projection theorem directly. The package is now named `Theorem49BoundaryRawQuotientErrorPackage`, and the current replacement positive geometry surfaces consume it directly. The fixed-embedding theorems `theorem49BoundaryRawQuotientErrorPackage_of_heightOrderedPositiveProjectedGeometryOn` and `theorem49BoundaryRawQuotientErrorPackage_of_collarLayerPositiveProjectedGeometryOn` thread the endpoint through height-ordered and finite collar-layer witness-cover data, and the graph-level methods on `Theorem49HeightOrderedPositiveProjectedGeometry` and `Theorem49CollarLayerPositiveProjectedGeometry` expose the same conclusion on the packaged embedding. This is the current non-vacuous replacement route's raw quotient/error consumption point. At the core synthesis surface this has been strengthened to a submodule equality: `Theorem49BoundaryRootSynthesis.kirchhoffSubmodule_eq_rawSource_sup_boundaryKernel`. It identifies the purified Kirchhoff submodule with the supremum of the Kirchhoff-valid raw Definition 4.8 Kempe-source submodule and the selected-boundary projection kernel error submodule. Thus the quotient/error reading is no longer only a family of finite witnesses; Lean now records the global raw-source-plus-kernel decomposition forced by any completed `Theorem49BoundaryRootSynthesis`. The same synthesis layer now also records the quotient form `Theorem49BoundaryRootSynthesis.rawSource_mkQ_eq_top`: inside the purified Kirchhoff submodule, after quotienting by the kernel of the selected-boundary projection restricted to that submodule, the Kirchhoff-valid raw Definition 4.8 source maps onto the whole quotient. This is the exact surjectivity statement needed by the quotient/error interpretation, and it avoids replacing the purified Kirchhoff quotient by a stronger ambient-space quotient. The theorem `Theorem49BoundaryRootSynthesis.finrank_quotient_le_rawSource` records the corresponding finite-dimensional consequence: the purified Kirchhoff quotient by the restricted selected-boundary kernel has finrank no larger than the raw Definition 4.8 source submodule that surjects onto it. The target-side bound `Theorem49BoundaryRootSynthesis.finrank_target_le_rawSource` also lives at the same surface: the concrete boundary-zero Kirchhoff target has finrank no larger than the Kirchhoff-valid raw Definition 4.8 source whose projection is that target. The actual quotient-target identification is now available as `Theorem49BoundaryRootSynthesis.boundaryProjectionQuotientEquivTarget`, with `Theorem49BoundaryRootSynthesis.boundaryProjectionQuotientEquivTarget_apply_mk` recording that a quotient class represented by a purified Kirchhoff chain maps to its selected-boundary-erased chain in the concrete target. The raw-source quotient version `Theorem49BoundaryRootSynthesis.rawSourceQuotientEquivTarget` now identifies the Kirchhoff-valid raw Definition 4.8 source quotient, modulo the selected-boundary projection kernel on that source, with the same concrete target; its `...rawSourceQuotientEquivTarget_apply_mk` theorem gives the corresponding representative computation rule. The raw and purified quotient models are now connected by `Theorem49BoundaryRootSynthesis.rawSourceQuotientEquivKirchhoffQuotient`. This linearly identifies the raw-source quotient with the purified Kirchhoff quotient through their common concrete theorem-4.9 target, and `Theorem49BoundaryRootSynthesis.rawSourceQuotientEquivKirchhoffQuotient_apply_mk` records the representative computation: a raw Kirchhoff-valid source class maps to the same edge-chain class in the purified Kirchhoff quotient. This is a genuine comparison map between the quotient/error surfaces, not another package threading variant. The same synthesis layer now also records the finite representative form `Theorem49BoundaryRootSynthesis.exists_rawGeneratorSum_for_kirchhoffQuotientClass`: each purified Kirchhoff quotient class admits a representative lying in the raw Definition 4.8 source and given as a finite linear combination of raw generator-family terms. The conclusion already includes equality in the purified Kirchhoff quotient, so the selected-boundary projection kernel conversion is discharged once at the synthesis surface. The quotient/error surface now carries the separation test as well: `Theorem49BoundaryRootSynthesis.kirchhoffQuotientClass_eq_zero_iff_projectedGenerator_orthogonal` identifies zero purified Kirchhoff quotient classes with classes whose selected-boundary-erased representatives are orthogonal to all projected Definition 4.8 generators. Its detector companion `Theorem49BoundaryRootSynthesis.exists_projectedGenerator_chainDot_ne_zero_of_ne_zero_kirchhoffQuotientClass` extracts a projected generator with nonzero chain-dot pairing from any nonzero purified quotient class. This makes projected-generator separation usable directly modulo selected-boundary kernel error. The combined theorem `Theorem49BoundaryRootSynthesis.exists_rawGeneratorSum_and_projectedGenerator_chainDot_ne_zero_of_ne_zero_kirchhoffQuotientClass` now gives, for every nonzero purified quotient class, a finite raw Definition 4.8 generator-sum representative together with a projected generator that detects that representative after selected-boundary erasure. This is the usable synthesis-surface form for nonzero quotient classes. The quotient surface also now has a named linear-map form: `Theorem49BoundaryRootSynthesis.kirchhoffQuotientProjectedGeneratorPairingMap` pairs purified Kirchhoff quotient classes with projected Definition 4.8 generators after selected-boundary erasure. Its `..._apply_mk` theorem computes the pairing on representatives, and `Theorem49BoundaryRootSynthesis.ker_kirchhoffQuotientProjectedGeneratorPairingMap_eq_bot` records the quotient-level trivial-kernel statement. `Theorem49BoundaryRootSynthesis.kirchhoffQuotientProjectedGeneratorPairingMap_eq_zero_iff` and `Theorem49BoundaryRootSynthesis.exists_projectedGenerator_pairingMap_ne_zero_of_ne_zero_kirchhoffQuotientClass` turn that kernel statement into the map-level detector for arbitrary purified quotient classes. `Theorem49BoundaryRootSynthesis.finrank_kirchhoffQuotient_le_projectedGeneratorSubspace` then extracts the corresponding finite-dimensional bound for the purified Kirchhoff quotient from that named pairing map, rather than appealing only to the older target-side finrank field, and `Theorem49BoundaryRootSynthesis.finrank_range_kirchhoffQuotientProjectedGeneratorPairingMap` identifies the pairing-map image dimension with the purified quotient dimension. The new equivalence `Theorem49BoundaryRootSynthesis.kirchhoffQuotientEquivPairingMapRange` then uses this same injective map to model purified quotient classes as functionals in the pairing-map image, with `Theorem49BoundaryRootSynthesis.kirchhoffQuotientEquivPairingMapRange_apply_mk` retaining the representative-level chain-dot computation.

`Theorem49BoundaryRootSynthesis.rawSourceQuotientEquivPairingMapRange` now gives the parallel image model for the raw Definition 4.8 source quotient itself, and `Theorem49BoundaryRootSynthesis.rawSourceQuotientEquivPairingMapRange_apply_mk` computes raw-source quotient representatives by selected-boundary erasure and chain-dot pairing. The raw-source quotient also has its own projected-generator separation test: `Theorem49BoundaryRootSynthesis.rawSourceQuotientClass_eq_zero_iff_projectedGenerator_orthogonal` and `Theorem49BoundaryRootSynthesis.exists_projectedGenerator_chainDot_ne_zero_of_ne_zero_rawSourceQuotientClass` characterize zero raw-source classes and detect nonzero ones without leaving the quotient/error surface. The finite representative theorem `Theorem49BoundaryRootSynthesis.exists_rawGeneratorSum_and_projectedGenerator_chainDot_ne_zero_of_ne_zero_rawSourceQuotientClass` keeps the same raw-source quotient class while choosing an explicit finite raw Definition 4.8 generator-family sum and a projected generator that detects it. The concrete target-level theorem `Theorem49BoundaryRootSynthesis.exists_rawGeneratorSum_and_projectedGenerator_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace` now combines the raw projection representation and pointwise separation: every nonzero boundary-zero Kirchhoff target has a finite raw generator-family preimage and a projected generator that detects that same target. The sharpened theorem `Theorem49BoundaryRootSynthesis.exists_nonempty_rawGeneratorSum_and_nonzero_projectedGenerator_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace` records the non-vacuity content of the same conclusion: the finite raw generator-family representative is not the empty sum, and the projected detector is itself nonzero. The strengthened support-sensitive theorem `Theorem49BoundaryRootSynthesis.exists_nonzero_rawGeneratorSum_and_nonempty_coeffSupport_and_nonzero_projectedGenerator_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace` adds that this raw representative is a nonzero chain and that the coefficient support is nonempty, not only the ambient term finset. The term-level witness theorem `Theorem49BoundaryRootSynthesis.exists_rawGenerator_with_nonzero_coeff_and_nonzero_projectedGenerator_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace` extracts a concrete raw Definition 4.8 generator-family chain from that support with nonzero coefficient, while keeping the same nonzero raw representative and projected detector data. `Theorem49BoundaryRootSynthesis.exists_nonzero_rawGenerator_with_nonzero_coeff_and_nonzero_projectedGenerator_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace` removes the last degeneracy in this witness: the extracted support chain can be chosen nonzero, so the target-level detector is now tied to an actual nonzero raw Definition 4.8 generator-family chain. `Theorem49BoundaryRootSynthesis.exists_rawGenerator_with_nonzero_boundaryZeroProjection_and_nonzero_coeff_and_nonzero_projectedGenerator_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace` adds the boundary-facing non-vacuity condition: the extracted support chain can be chosen with nonzero selected-boundary erasure, so a nonzero theorem-4.9 target is not represented only by support terms killed by boundaryZeroProjection. The raw-source quotient now also has the rank consequences of that image model: `Theorem49BoundaryRootSynthesis.finrank_rawSourceQuotient_eq_pairingMapRange` identifies its dimension with the pairing-map image dimension, and `Theorem49BoundaryRootSynthesis.finrank_rawSourceQuotient_le_projectedGeneratorSubspace` bounds it by the projected Definition 4.8 generator subspace. This separation layer has now been lowered through

`PlanarBoundaryClosedWalkSourceProjection.lean` to the actual source shells: canonical witness choice, the local at-most-one criterion, and same-boundary one-collar geometry, for both closed-walk sources and successor-cycle boundary-order sources, give the same chain-dot separation endpoint through height-ordered and finite collar-layer witness data. The source file now lowers the matching pointwise detector and orthogonality iff forms too, so these same source shells can test each nonzero boundary-zero Kirchhoff chain by a projected Definition 4.8 generator; it now also lowers the raw preimage and finite raw-generator sum forms, so each such chain has a raw Definition 4.8 generator-span representative and a finite raw-generator combination projecting to it. The positive non-vacuous lane now has a single same-embedding package, `Theorem49BoundaryRootNonemptyProjectedSynthesis`, combining `Theorem49BoundaryRootNonemptySynthesis` with the corrected projected subspace and projected-family span equalities. Height-ordered and finite collar-layer witness data construct the package under nonempty purified carrier, and the closed-walk and successor-cycle source wrappers expose it for canonical witness choice and same-boundary one-collar geometry. The at-most-one/nonempty-carrier branch is intentionally absent from this positive package because the current purified carrier makes that branch inconsistent;

- `Theorem49PositiveGeometricSource.lean` names the candidate geometric source packages that were isolated for testing: `Theorem49ClosedWalkCanonicalPositiveGeometry`, `Theorem49ClosedWalkOneCollarPositiveGeometry`, `Theorem49SuccessorCycleCanonicalPositiveGeome` and `Theorem49SuccessorCycleOneCollarPositiveGeometry`. These packages keep the source data, canonical witness choice or source-preserving one-collar geometry, and nonempty purified carrier on the same embedding, then lower to `Theorem49BoundaryRootNonemptyProjectedSynthesis` for any Tait coloring. The fixed-embedding predicate forms now lower directly to the same endpoint on the same embedding;

- `Theorem49PositiveGeometricSourceRegression.lean` supplies that focused obstruction for the wheel-with-inner-triangle benchmark. The fixed embedding retains `wheelWithInnerTriangleTaitEdgeColoring_isTait` and `selectedBoundaryInteriorEdgeEndpointVertices_nonempty_wheelWithInnerTriangle`, but Lean proves the four package-level failures `not_theorem49ClosedWalkCanonicalPositiveGeometryOn_wheelWithInnerTriangle`, `not_theorem49ClosedWalkOneCollarPositiveGeometryOn_wheelWithInnerTriangle`, `not_theorem49SuccessorCycleCanonicalPositiveGeometryOn_wheelWithInnerTriangle`, and `not_theorem49SuccessorCycleOneCollarPositiveGeometryOn_wheelWithInnerTriangle`. Thus the wheel benchmark is no longer just an informal separator between carrier and geometry: it is a fixed-embedding regression test for the current positive package layer;

- `Theorem49PositiveGeometricSourceImpossibility.lean` upgrades that benchmark calibration to a structural route correction. Canonical witness choice implies facewise at-most-one interior edge, and honest closed-walk data plus facewise at-most-one empties the purified carrier. Same-boundary one-collar geometry induces canonical witness choice, so `not_nonempty_currentTheorem49PositiveGeometricSourcePackages` refutes all four current graph-level positive package structures. The boundary-order next step is therefore no longer to instantiate these packages, but to replace the source surface or carrier condition so that the nonempty projected endpoint is not forced empty. The same obstruction now has direct live-endpoint forms: `not_hasUnblockedInteriorEndpoint_` of_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice and not_hasUnblockedInteriorEndpoint_of_closedWalkAnnulusBoundarySource_and_oneCollarGeometry_with_sourceBoundaryData say that canonical choice and source-preserving one-collar geometry rule out the local endpoint witness itself. Equivalently, not_nonempty_planarBoundaryCanonicalWitnessChoice_of_closedWalkAnnulusBoundarySource_and_hasUnblockedInteriorEndpoint blocks canonical choice from an honest source plus HasUnblockedInteriorEndpoint, and not_exists_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_of_closedWalkAnnulusBoundarySource_and_hasUnblockedInteriorEndpoint blocks same-boundary one-collar geometry;

- Theorem49PositiveGeometricSourceReplacement.lean now records the direct replacement surface. The fixed-embedding predicates Theorem49HeightOrderedPositiveProjectedGeometryOn and Theorem49CollarLayerPositiveProjectedGeometryOn pair the corresponding boundary-aware witness-cover data with nonempty purified carrier on the same embedding and lower directly to Theorem49BoundaryRootNonemptyProjectedSynthesis. The graph-level packages Theorem49HeightOrderedPositiveProjectedGeometry and Theorem49CollarLayerPositiveProjectedGeometry expose the same endpoint, and the theorem heightOrderedPositiveProjectedGeometryOn_iff_collarLayerPositiveProjectedGeometryOn identifies the two fixed-embedding forms using the existing height/collar conversions. The construction task has therefore been sharpened to building one of these direct witness-data packages from boundary-order geometry, while the file's compatibility lemmas prevent silently reintroducing canonical-choice or source-preserving one-collar fields on a nonempty carrier embedding. The same file now exposes the graph-level annulus bridge explicitly: theorem49CollarLayerPositiveProjectedGeometry_of_annulusCollarGeometry and theorem49HeightOrderedPositiveProjectedGeometry_of_annulusCollarGeometry turn annulus collar geometry plus nonempty purified carrier into the replacement packages, with existential versions and repaired previous-boundary-witness specializations. The same minimal hypotheses now also reach the nonempty projected synthesis endpoint directly via exists_theorem49BoundaryRootNonemptyProjectedSynthesis_of_exists_annulusCollarGeometry_and_nonempty_selectedBoundaryInteriorEdgeEndpointVertices_direct. The repaired previous-boundary-witness lane now has the endpoint-witness version as well: theorem49BoundaryRootNonemptyProjectedSynthesis_of_annulusPreviousBoundaryWitnessGeometry_and_hasUnblockedInteriorEndpoint uses HasUnblockedInteriorEndpoint directly, and the route file adds the closed-walk source-facing bridge theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySource_and_annulusPreviousBoundaryWitnessGeometry_and_hasUnblockedInteriorEndpoint. The acyclic witness-cover lane now has the same direct endpoint: theorem49BoundaryRootNonemptyProjectedSynthesis_of_wellFoundedFacePeelWitnessData_and_hasUnblockedInteriorEndpoint uses well-founded parent witness-cover data plus HasUnblockedInteriorEndpoint to reach the projected synthesis surface, and the source-facing route theorem theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySource_and_wellFoundedFacePeelWitnessData_and_hasUnblockedInteriorEndpoint keeps the honest closed-walk source visible while making the acyclic data the actual geometric input. This direct acyclic lane now also consumes endpoint-support separation and raw endpoint-carrier nonemptiness without first manufacturing the named endpoint witness. The generic theorem `theorem49BoundaryRootNonemptyProjectedSynthesis_of_wellFoundedFacePeelWitnessData_and_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport` has height-ordered, finite-collar, graph-level, and existential variants, while the source-facing wrappers `theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySource_and_wellFoundedFacePeelWitnessData_and_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport` and `theorem49BoundaryRootNonemptyProjectedSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_wellFoundedFacePeelWitnessData_and_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport` show that honest closed-walk and successor-cycle source shells can use the same endpoint-support bridge once explicit well-founded witness-cover data is available. This is a packaging advance, not a derivation of the acyclic witness-cover datum itself. The root-distance interior-dual route now produces that acyclic input on the same embedding: `planarBoundaryWellFoundedFacePeelWitnessData_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData` turns honest source data plus generic boundary-root interior-dual adjacency-distance data into explicit well-founded witness-cover data, and `planarBoundaryHeightOrderedFacePeelWitnessData_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData` turns the same root-distance parent order into the height-ordered witness-cover surface directly. The graph-level source theorem is `admitsPlanarBoundaryHeightOrderedFacePeelWitnessData_of_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_direct`. The obstruction side has also been factored at this same witness-cover interface. In `Theorem49PlanarBoundaryPeeling.lean`, a non-boundary remainder edge whose value is another peeled face's witness edge forces a parent edge in the well-founded package, or a strict height inequality in the height-ordered package, provided the witness choice is injective on peeled faces. The resulting two- and three-cycle lemmas rule out directed cycles of these non-boundary remainder-witness dependencies before any wheel-specific argument is used. This has now been generalized to arbitrary nonempty finite closed dependency chains through `false_of_isChain_height_lt_and_getLast_lt_head` and the well-founded and height-ordered `false_of_witnessRemainderDependency_cycle_of_injective_witness` lemmas. The wheel-with-inner-triangle regression now exercises that list-level lemma directly: the two possible orientations of the height-ordered spoke chase are closed as explicit three-face finite cycles, not as separate ad hoc irreflexivity arguments. The well-founded parent side has also been separated from this height presentation. The certificate builder proves `false_of_isChain_parentRel_and_getLast_parent_head`, saying that a well-founded parent map has no nonempty finite closed `ParentRel` chain, and the witness-cover finite-cycle blocker now uses that parent-chain theorem after extracting the actual parent edges. Thus the minimal upstream lemma for deriving acyclic witness-cover data from raw closed-walk or successor-cycle boundary-order source hypotheses is now precise: construct `InteriorDualBoundaryRootAdjDistancePeelData` on the same embedding, or construct an equivalent cycle-breaking well-founded parent witness. The source-side primitive constructor `interiorDualBoundaryRootAdjDistancePeelDataOfClosedWalkAnnulusBoundarySourceCoverSeparated` now spells out the first option: starting from a `PlanarBoundaryClosedWalkAnnulusBoundarySource`, it is enough to supply cover/separated annulus-boundary roots, boundary-free peeled faces, canonical rooted shared-edge coverage of every interior edge, child-face membership with strict root-distance increase, and the ambient two-incidence bound. The graph-level theorem `admitsPlanarBoundaryInteriorDualBoundaryRootAdjDistancePeelData_of_exists_closedWalkAnnulusBoundarySource_and_coverSeparatedAnnulusBoundaryRootsCanonicalRootedShare` records the same source-side obligation. The refined constructor `interiorDualBoundaryRootAdjDistancePeelDataOfClosedWalkAnnulusBoundarySourceBoundaryFaceRo` then fixes the roots to the source's outer-or-inner boundary faces, fixes the fallback edge to the honest closed-walk fallback, and uses the embedding's two-incidence theorem. Consequently the live source-side obligation is cover/separation of the source boundary-face roots, boundary-free peeled faces, rooted shared-edge coverage of the interior support, and strict root-distance child membership. The parent-oriented sufficiency option is now formalized: `PlanarBoundaryClosedWalkAnnulusBoundarySource.wellFounded_parentRel_boundaryFaceRoots_parentTowardsRoot` proves the acyclic orientation part of the sufficiency direction: once the source boundary-face roots cover and separate the interior-dual components, the canonical shortest-path parent map toward those roots is well founded. Lean also packages the parent-oriented fixed-root route as `interiorDualBoundaryRootParentPeelDataOfClosedWalkAnnulusBoundarySourceBoundaryFaceRootsCa` and its graph-level theorem `admitsPlanarBoundaryInteriorDualBoundaryRootParentPeelData_of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover`. Thus the source-fixed sufficiency target can now be attacked as IDBRAD from the raw rooted shared-edge cover; canonical parent child-membership is derived internally by `canonicalParentChildMembership_of_canonicalParentCover`. The older `admitsPlanarBoundaryInteriorDualBoundaryRootParentPeelData_of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov` theorem remains a compatibility endpoint for callers already carrying the derived child-membership field. The same layer now has a proved empty-interior-edge base case: `interiorDualBoundaryRootParentPeelDataOfClosedWalkAnnulusBoundarySourceBoundaryFaceRootsNo` and `admitsPlanarBoundaryInteriorDualBoundaryRootParentPeelData_of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsNoInteriorEdges` construct the source-fixed parent package with `peelFaces=\emptyset` whenever the source boundary-face roots cover and separate and `interiorEdgeSupport=\emptyset`. Hence the nondegenerate burden is now precisely the pairwise-unique shared-edge selector input, root cover/separation, boundary-free peeled faces, and rooted shared-edge coverage of actual interior edges. The empty-support case obtains the required shared-edge uniqueness internally from `pairwiseUniqueSharedInteriorEdges_of_interiorEdgeSupport_eq_empty`, so it does not assume pairwise uniqueness as an independent source hypothesis. The nonempty-support branch now has a constructor-side witness theorem instead of only selector-level consequences. The generic lemma `exists_nonroot_peelFace_parent_rootedSharedInteriorEdge_eq_of_canonicalParentCover_of_mem_interiorEdgeSupport` shows that any actual interior edge covered by the canonical rooted shared-edge image is covered by a peeled non-root face with a real shortest-path parent, and that the rooted shared edge on that face is exactly the original interior edge. Its nonempty-support version `exists_nonroot_peelFace_parent_rootedSharedInteriorEdge_eq_of_canonicalParentCover_of_interiorEdgeSupport_nonempty` makes the nondegenerate forest branch explicit. The closed-walk source specializations `exists_nonroot_peelFace_parent_rootedSharedInteriorEdge_eq_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov of_mem_interiorEdgeSupport` and `exists_nonroot_peelFace_parent_` rootedSharedInteriorEdge_eq_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalPa of_interiorEdgeSupport_nonempty instantiate the witness with the source boundary-face roots and source fallback edge. Therefore a nonempty canonical-parent cover cannot hide in the fallback case. The constructor layer now also has parent-side membership for canonical shared edges and canonicalParentChildMembership_of_canonicalParentCover, which turns the raw cover plus the two-incidence bound into the parent child-membership field. The remaining hard source task is the actual boundary-order proof of the pairwise-unique shared-edge selector input, root cover/separation, boundary-free peeled faces, and rooted shared-edge cover. The nondegenerate raw-cover constructor has now been tested directly on the larger chained-diamonds honest source. The regression chainedDiamondsWithTriangle_interiorEdgeSupport_nonempty records genuine interior support, and chainedDiamondsWithTriangleForcingInteriorEdgeWitness packages cdt12 as an interior edge all of whose incident faces already touch selected-boundary support. The theorem false_of_chainedDiamondsClosedWalkSourceBoundaryFaceRootsCanonicalParentSharedEdgeCover then calls interiorDualBoundaryRootParentPeelDataOfClosedWalkAnnulusBoundarySourceBoundaryFac on the canonical chained-diamonds source and derives a contradiction from the forcing-edge obstruction. Thus this concrete nonempty-interior closed-walk benchmark cannot instantiate the source-fixed raw parent constructor: the missing datum is a genuine boundary-free incident peeled face, not another downstream selector or quotient wrapper. The obstruction is now source-level rather than merely benchmark-specific. PlaneEmbeddingWithBoundary.FaceBoundaryClosedWalk.three_le_faceBoundary_card proves that every honest nonempty simple facial closed walk has at least three boundary edges. Combining that lower bound with the terminal-face theorem for PlanarBoundaryHeightOrderedFacePeelWitnessData, Lean proves false_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_of_interiorEdgeSupport_nonempty. Indeed, the raw source-fixed constructor lowers to a height-ordered witness cover; any nonempty interior support gives a terminal peeled face whose non-witness boundary edges already lie in selected-boundary support. Since the raw cover also requires peeled faces to be disjoint from selected-boundary support, the terminal face boundary is contained in the singleton witness edge, contradicting the closed-walk lower bound. Therefore the current honest closed-walk source plus boundary-free peeled-face interface has no positive nondegenerate raw-cover instantiation; the sufficiency route must change this source-side contract or replace the terminal parent condition before a genuine nonempty benchmark can exist. The degeneracy is also exposed as a usable route theorem: interiorEdgeSupport_eq_empty_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover concludes interiorEdgeSupport=\emptyset from the same raw cover hypotheses. This makes the existing no-interior constructor the only possible honest closed-walk source-fixed instance of that raw-cover contract, rather than merely a base case next to an open nonempty branch. The obstruction has also been lifted to the adjacent-distance/root-parent interface itself. The theorem InteriorDualBoundaryRootAdjDistancePeelData.false_of_mem_interiorEdgeSupport_of_faceBoundaryClosedWalkGeometry instantiates the existing terminal-card obstruction from PlaneEmbeddingWithBoundary.FaceBoundaryClosedWalkGeometry: a covered interior edge would force a peeled face with at most one boundary edge, but the honest closed-walk lower bound gives at least three boundary edges on every face. Consequently not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_of_interiorEdgeSupport_

`nonempty_of_faceBoundaryClosedWalkGeometry` rules out any IDBRAD package on an honest closed-walk embedding with genuine interior support. Thus the defect is not merely a bad source-fixed raw-cover constructor; the current boundary-free terminal peeling interface must be changed before it can support a nondegenerate sufficiency route. This base case is now instantiated concretely in `Theorem49PlanarBoundaryAnnulusRootInteriorDualPositiveRegression.lean`. The finite witness `twoTriangleAnnulusGraph` has two disjoint triangular closed-walk boundary components, source boundary-face roots equal to all faces by `twoTriangleAnnulusBoundaryFaceRoots_eq_univ`, an empty interior dual, and empty `interiorEdgeSupport`. Consequently `twoTriangleClosedWalkSourceInteriorDualBoundaryRootParentPeelData` and `admitsPlanarBoundaryInteriorDualBoundaryRootParentPeelData_twoTriangleAnnulusGraph` are genuine positive source-fixed sufficiency checks, complementary to the two-band obstruction where separation and boundary-free cover fail. The same base case now reaches the selector surface used by the replacement-construction lane: `boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData` extracts a `BoundaryFreeIncidentFaceSelector` from any canonical-parent payload while retaining the parent cover data: `boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData_faceOf_mem_peelFaces` keeps every selected face inside the parent package's `peelFaces`, and `rootedSharedInteriorEdge_boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData_faceOf_eq` identifies the selected face's rooted shared edge with the original interior edge. The construction-layer compatibility theorem `boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData_peelFaces_subset` shows that the selector's finite image introduces no extra peeled faces. `boundaryFreeIncidentFaceSelectorOfClosedWalkAnnulusBoundarySourceBoundaryFaceRootsNoInteri` is the source-fixed empty-support specialization, and `twoTriangleClosedWalkSourceBoundaryFreeIncident` instantiates that selector on the concrete two-triangle source. The bridge is now strict-descent compatible as well: `boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData_injective` proves injectivity on interior edges, `boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData_selectedWitnessEdge_eq` identifies the selector's inverted witness with the canonical rooted shared edge, and `boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData_strictDescent` turns parent child-membership into selector `hrest` using root distance. The route theorem `theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov`
`via_boundaryFreeSelector` therefore composes the source-fixed parent hypotheses through the selector-to-construction lane directly. The selector path now also has the route-facing raw quotient/error corollary `theorem49BoundaryRawQuotientErrorPackage_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov`
`via_boundaryFreeSelector` and the graph-level endpoint wrapper `exists_theorem49BoundaryRootNonemptyProjectedSynthesis_of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov`
`and_hasUnblockedInteriorEndpoint_via_boundaryFreeSelector`, where the carrier side is the local `HasUnblockedInteriorEndpoint` witness. The parent-oriented source packet now also reaches the route endpoint: `theorem49BoundaryRootNonemptyProjectedSynthesis_of_`

closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov direct packages the fixed source, source boundary-face roots, canonical parent child-membership, and a nonempty purified carrier into Theorem49BoundaryRootNonemptyProjectedSynthesis; the graph-level counterpart is exists_theorem49BoundaryRootNonemptyProjectedSynthesis_of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov with_nonempty_selectedBoundaryInteriorEdgeEndpointVertices_direct. The wheel-with-inner-triangle regression now instantiates this source-fixed parent bridge at the actual endpoint: wheelWithInnerTriangle_theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkSourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover specializes the direct theorem to the concrete wheel embedding and its explicit Tait coloring. The companion theorem wheelWithInnerTriangle_source_to_theorem49_and_height_cycle_contradiction_of_closedWalkSourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover lowers the same parent hypotheses back to height-ordered witness-cover data and therefore contradicts the existing wheel height-cycle obstruction. Thus not_exists_wheelWithInnerTriangle_closedWalkSourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_and_theorem49BoundaryRootNonemptyProjectedSynthesis is the endpoint-level soundness check for this parent route: the wheel keeps its honest source, Tait coloring, and unblocked endpoint, but cannot instantiate the source-fixed canonical-parent hypotheses. The coverage sub-obligation is now separately accounted for: PlanarBoundaryClosedWalkAnnulusBoundarySource.rootSetCovers_boundaryFaceRoots_of_interiorDualBoundaryRootAdjDistancePeelData proves that any generic boundary-root IDBRAD datum already makes the source boundary-face roots cover all interior-dual components. The separation sub-obligation is not automatic: PlanarBoundaryClosedWalkAnnulusBoundarySource.not_rootSetSeparated_boundaryFaceRoots_of_reachable_outerBoundaryFace_innerBoundaryFace shows that an outer-to-inner interior-dual reachability path refutes separated source boundary-face roots. The stronger source-side obstruction PlanarBoundaryClosedWalkAnnulusBoundarySource.not_reachable_outerBoundaryFace_innerBoundaryFace_of_interiorDualBoundaryRootAdjDistancePeelData shows that any generic IDBRAD datum on the closed-walk source already rules out such a path, and PlanarBoundaryClosedWalkAnnulusBoundarySource.not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_of_reachable_outerBoundaryFace_innerBoundaryFace packages this as nonexistence of the generic datum under outer-to-inner interior-dual reachability. The same obstruction is now exposed in route-facing form by not_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_and_reachable_outerBoundaryFace_innerBoundaryFace and not_forall_interiorDualBoundaryRootAdjDistancePeelData_of_exists_closedWalkAnnulusBoundarySource_and_reachable_outerBoundaryFace_innerBoundaryFace. This is now a concrete regression, not just an abstract no-coexistence theorem. Theorem49PlanarBoundaryAnnulusRootInteriorDualObstructionRegression.lean defines the two-band triangular annulus twoBandAnnulusGraph, proves its closed-walk source twoBandClosedWalkAnnulusBoundarySource, and proves that the selected outer face 0 and selected inner face 3 are adjacent in the interior dual through the middle edge tbaM01. Consequently not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_ twoBandAnnulus_via_source_reachability instantiates the source-side IDBRAD obstruction on a finite graph with an honest closed-walk annulus source. The same concrete source now proves the fixed-root calibration directly: twoBandAnnulusBoundaryFaceRoots_eq_univ identifies the source boundaryFaceRoots with all six faces, rootSetCovers_twoBandAnnulusBoundaryFaceRoots proves coverage, and not_rootSetSeparated_twoBandAnnulusBoundaryFaceRoots proves separation false using the shared middle edge tbaM01. Any future derivation from the source shell must therefore add or prove an actual outer/inner face-separation condition. The same regression now proves the lower-level peel obstruction twoBandAnnulus_peelFaces_eq_empty_of_boundaryFree: every face touches selected boundary support, so a boundary-free peel-face set is empty. Since tbaM01 is still an interior edge, not_interiorEdgeSupport_subset_boundaryFreePeelImage_twoBandAnnulus rules out an interior-edge-support cover by any image of those faces. The same obstruction is now registered in the common forcing-edge interface: twoBandAnnulusForcingInteriorEdgeWitness packages tbaM01, not_nonempty_boundaryFreeIncidentFaceSelector_twoBandAnnulus rules out the selector surface, and not_nonempty_planarBoundaryAnnulusRootAdjDistancePeelData_twoBandAnnulus_via_forcingInteriorEdgeWitness rules out the annulus-root adjacency-distance package on this embedding. The same forcing-edge guard now covers the parent-oriented sufficiency target: everyInteriorEdgeHasBoundaryFreeIncidentFace_of_interiorDualBoundaryRootParentPeelData extracts a boundary-free incident face from any canonical-parent boundary-root package, while not_nonempty_interiorDualBoundaryRootParentPeelData_twoBandAnnulus_via_forcingInteriorEdgeWitness and not_nonempty_planarBoundaryAnnulusRootParentPeelData_twoBandAnnulus_via_forcingInteriorEdgeWitness instantiate the failure on the same two-band source. The generic IDBRAD layer now has a source-side terminal obstruction independent of the concrete annuli. In Theorem49PlanarBoundaryPeeling.lean, PlanarBoundaryHeightOrderedFacePeelWitnessData.exists_terminal_face_selectedBoundary_remainders shows that any nonempty height-ordered witness cover has a terminal peeled face whose non-witness remainders are all selected-boundary edges. The IDBRAD boundary-free peeled-face invariant then forces those remainders to be empty: InteriorDualBoundaryRootAdjDistancePeelData.exists_terminal_peelFace_witness_erase_eq_empty_of_mem_interiorEdgeSupport. Consequently InteriorDualBoundaryRootAdjDistancePeelData.exists_peelFace_card_le_one_of_mem_interiorEdgeSupport says that any IDBRAD datum covering even one interior edge contains a peeled face of boundary cardinality at most one. Future source-side derivations must therefore produce this terminal singleton-edge face, or replace IDBRAD by a different witness-ownership surface. The same criterion is now instantiated on the wheel-with-inner-triangle source without passing through the forcing-edge API. The regression proves wheelWithInnerTriangleFaceBoundary_card_eq_three and wheelWithInnerTriangle_two_le_faceBoundary_card; together with wit01_mem_interiorEdgeSupport, these give not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_wheelWithInnerTriangle_via_terminal_card_lower_bound. The packaged source bridge wheelWithInnerTriangle_closedWalkSource_tait_hasUnblockedEndpoint_without_interiorDualBoundaryRootAdjDistancePeelData_via_terminal_card keeps the honest closed-walk source, Tait coloring, and local unblocked endpoint while deriving IDBRAD nonexistence from the terminal-card lower-bound obstruction. This is now tied to the real projected Theorem 4.9 endpoint on the same benchmark. The wheel file supplies wheelWithInnerTriangleGraph_edgeSet_fintype, and the regression names hasUnblockedInteriorEndpoint_wheelWithInnerTriangle. With those facts, wheelWithInnerTriangle_theorem49BoundaryRootNonemptyProjectedSynthesis_of_interiorDualBoundaryRootAdjDistancePeelData instantiates the closed-walk source–IDBRAD bridge for the concrete wheel Tait coloring, and wheelWithInnerTriangle_theorem49BoundaryRawQuotientErrorPackage_of_interiorDualBoundaryRootAdjDistancePeelData instantiates the raw Definition 4.8 quotient/error endpoint. The theorem wheelWithInnerTriangle_source_to_theorem49_and_terminal_card_contradiction_of_interiorDualBoundaryRootAdjDistancePeelData then records that the same hypothetical IDBRAD datum would also contradict the terminal-card obstruction. The endpoint-level form is now explicit: not_exists_wheelWithInnerTriangle_closedWalkSource_and_interiorDualBoundaryRootAdjDistancePeelData_and_theorem49BoundaryRootNonemptyProjectedSynthesis rules out coexistence of the closed-walk source, same-embedding IDBRAD datum, and projected Theorem 4.9 endpoint on the wheel. The packaged positive-source statement wheelWithInnerTriangle_closedWalkSource_tait_hasUnblockedEndpoint_without_interiorDualBoundaryRootAdjDistancePeelData_based_projectedSynthesis retains the honest source, Tait coloring, and unblocked endpoint while ruling out an IDBRAD-based projected endpoint instantiation. The same regression now performs the analogous check for the source-fixed canonical-parent route: wheelWithInnerTriangle_closedWalkSource_tait_hasUnblockedEndpoint_without_boundaryFaceRootsCanonicalParent_based_projectedSynthesis keeps those positive source facts while ruling out a projected endpoint obtained from the source boundary-face roots and canonical parent child-membership. With the resulting acyclic datum in hand, theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_and_hasUnblockedInteriorEndpoint reaches the projected endpoint once the local endpoint witness is supplied. The successor-cycle source version is theorem49BoundaryRootNonemptyProjectedSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_interiorDualBoundaryRootAdjDistancePeelData_and_hasUnblockedInteriorEndpoint. The route file now also exposes the raw quotient/error form of this same source-side bridge: theorem49BoundaryRawQuotientErrorPackage_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_and_hasUnblockedInteriorEndpoint and theorem49BoundaryRawQuotientErrorPackage_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_interiorDualBoundaryRootAdjDistancePeelData_and_hasUnblockedInteriorEndpoint return the raw Definition 4.8 representative plus selected-boundary-kernel decomposition for any Kirchhoff chain on the same embedding. The same root-distance source layer now also accepts the older carrier-side repairs directly: endpoint-support separation plus raw interiorEdgeEndpointSupport nonemptiness, or peeled-face endpoint no-touch on planarBoundaryAnnulusConstruction_of_boundaryReachabilityData_and_interiorDualBoundaryRootAdjDistancePeelData_rootDistance plus the same raw nonemptiness. Thus the source/interior-dual projected endpoint no longer has to be called only with a prepackaged HasUnblockedInteriorEndpoint; it can consume the more geometric separation or root-distance no-touch obligations itself. That raw carrier can now be supplied at the edge-support level:

`interiorEdgeEndpointSupport_nonempty_of_interiorEdgeSupport_nonempty` turns nonempty `interiorEdgeSupport` into nonempty raw endpoint support, and the closed-walk root-distance bridge has direct projected-synthesis variants for both endpoint-support separation and root-distance peeled-face no-touch with this edge-level carrier hypothesis. The same bridge family now also accepts `HasUnblockedInteriorEndpoint`, with fixed-embedding, graph-level, nonempty-package, and projected-synthesis constructors from annulus collar geometry plus a local unblocked endpoint. Thus the remaining source-side burden is to derive those geometry-plus-carrier hypotheses, or directly the unblocked endpoint witness, not to repack the downstream endpoint;

- `Theorem49PositiveGeometricSourceReplacementRoute.lean` connects the replacement surface to both the honest closed-walk annulus source shell and the successor-cycle boundary-order source shell. It defines `Theorem49ClosedWalkAnnulusCollarPositiveProjectedGeometry`, which carries an honest source, same-embedding annulus collar geometry, and nonempty purified carrier, and it defines `Theorem49SuccessorCycleAnnulusCollarPositiveProjectedGeometry`, which carries boundary reachability, successor-cycle facial data, selected-boundary arcs, same-embedding annulus collar geometry, and nonempty purified carrier. The successor-cycle package lowers to the closed-walk route package for traceability; both packages lower to the replacement graph packages and directly to the nonempty projected synthesis endpoint. The constructive source-facing target is now explicit without canonical-choice or one-collar fields. The same route file also gives direct closed-walk and successor-cycle projected-synthesis constructors from `HasUnblockedInteriorEndpoint`, so future source examples can supply the carrier side by exhibiting one local endpoint disjoint from selected ambient-boundary support. The older endpoint-separation surface now supplies this witness directly: `hasUnblockedInteriorEndpoint_of_endpointSupport_disjoint_and_nonempty` turns raw `interiorEdgeEndpointSupport` nonemptiness plus disjointness from selected-boundary endpoint support into `HasUnblockedInteriorEndpoint`, and the route file has closed-walk and successor-cycle projected-synthesis wrappers for that raw endpoint-support form. The closed-walk and successor-cycle annulus-collar package constructors now also accept the same raw endpoint-support-separated input directly, so this is available before choosing a Tait coloring;

- the same endpoint-support bridge is now connected to the concrete collar geometry layer. The theorem `PlanarBoundaryAnnulusCollarGeometry.hasUnblockedInteriorEndpoint_of_peelFaces_endpoint_disjoint_selectedBoundarySupport` turns peeled-face endpoint no-touch on `planarBoundaryAnnulusConstruction_of_annulusCollarGeometrydata`, plus raw `interiorEdgeEndpointSupport` nonemptiness, into the named `HasUnblockedInteriorEndpoint` witness. The replacement positive package therefore has direct finite-collar, height-ordered, and projected-synthesis constructors from annulus collar geometry plus this construction-level no-touch surface. The route-facing replacement file now exposes the same input at the honest source layer: closed-walk and successor-cycle source data plus same-embedding annulus collar geometry, canonical peeled-face endpoint no-touch, and raw endpoint-support nonemptiness directly package the source-facing annulus-collar positive predicates and reach the nonempty projected synthesis endpoint. The same source-layer wrappers now also produce the graph-level finite-collar and height-ordered replacement packages directly, so the peeled-face no-touch input can be consumed by the replacement API without an intermediate route-package lowering step;

- `Theorem49PositiveGeometricSourceReplacementRegression.lean` calibrates the new direct target against the shared-interior-pair annulus. The same model has boundary reachability, successor-cycle selected arcs, a Tait coloring, and nonempty purified carrier, but it has neither `Theorem49HeightOrderedPositiveProjectedGeometryOn` nor `Theorem49CollarLayerPositiveProjectedGeometryOn`. The same file now also proves `not_theorem49ClosedWalkAnnulusCollarPositiveProjectedGeometryOn_sharedInteriorPair` and `not_theorem49SuccessorCycleAnnulusCollarPositiveProjectedGeometryOn_sharedInteriorPair`, so those facts do not universally construct either the direct replacement surface or the source-facing closed-walk/successor-cycle annulus-collar packages. The same regression now strengthens the carrier side to the local endpoint witness: `hasUnblockedInteriorEndpoint_sharedInteriorPair` and `not_forall_any_replacementPositiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` show that even explicit `HasUnblockedInteriorEndpoint` data, together with the same source and coloring fields, does not force any direct or route-facing replacement package. The closed-walk-source-level theorem `not_forall_any_replacementPositiveProjectedGeometryOn_of_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` strengthens this further: the honest closed-walk annulus source itself, a Tait coloring, and the endpoint witness still do not force any current direct or route-facing replacement package. The companion theorem `not_forall_some_witnessCoverData_of_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` rules out an even earlier shortcut: those same hypotheses also do not universally produce either raw height-ordered or collar-layer witness-cover datum. The bundled theorem `not_forall_some_currentSufficientSameEmbeddingGeometry_of_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` then rules out every current same-embedding synthesis endpoint from those hypotheses: height-ordered data, collar-layer data, the attached certificate, well-founded witness data, and annulus-collar geometry. A positive theorem must add real annulus-collar, height-ordered, or collar-layer witness-cover geometry. The wheel-with-inner-triangle regression now gives the complementary cycle obstruction at the honest-source level: `wheelWithInnerTriangleClosedWalkAnnulusBoundarySource` inhabits the closed-walk annulus source interface on the wheel embedding, while `not_forall_some_acyclicWitnessCover_or_annulusCollarGeometry_of_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_wheelWithInnerTriangle` shows that this source, a Tait coloring, and nonempty purified carrier still do not force well-founded, height-ordered, collar-layer, or weak annulus-collar geometry. The same strict height-cycle now reaches the direct replacement endpoints themselves: `not_theorem49HeightOrderedPositiveProjectedGeometryOn_wheelWithInnerTriangle` and `not_theorem49CollarLayerPositiveProjectedGeometryOn_wheelWithInnerTriangle` refute the two fixed-embedding positive replacement predicates, while `not_forall_some_replacementPositiveProjectedGeometry_or_previousBoundaryWitness_of_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_wheelWithInnerTriangle` also rules out deriving either direct replacement package or repaired previous-boundary witness geometry from just honest source data, Tait colorability, and the purified carrier. The generic cycle-obstruction lemmas in `Theorem49PlanarBoundaryPeeling.lean` explain the proof obligation exposed by this example: every proposed source-to-witness theorem must either put one of the remainder witnesses on selected boundary support, identify two witness choices so the apparent finite cycle collapses, or provide a genuine well-founded orientation;

- `Theorem49PositiveGeometricSourceConstruction.lean` gives the first positive construction of the direct replacement surface. On the two-collar annulus benchmark, Lean proves `counterEmbedding_selectedBoundaryInteriorEdgeEndpointVertices_nonempty` from the intermediate interior edge. `Theorem49PositiveGeometricSourceConstruction.lean` now also identifies the carrier exactly: `counterEmbedding_interiorEdgeSupport_eq` says the intermediate edge is the whole interior support, `counterEmbedding_interiorEdgeEndpointSupport_eq` and `counterEmbedding_selectedBoundaryEndpointSupport_eq` compute the raw interior endpoints and selected-boundary endpoints, and `counterEmbedding_endpointSupport_disjoint_selectedBoundarySupport` proves the disjointness that yields the named endpoint witness `counterEmbedding_hasUnblockedInteriorEndpoint`. This calibrates the benchmark against the first-order carrier obligation used by the replacement API. Lean then combines the carrier with annulus collar geometry to inhabit both `Theorem49CollarLayerPositiveProjectedGeometryOn` and `Theorem49HeightOrderedPositiveProjectedGeometryOn`, as well as the corresponding graph-level packages. The benchmark now also has an explicit constant nonzero Tait coloring, and Lean proves `counterEmbedding_boundaryRootNonemptyProjectedSynthesis` through the minimal annulus-collar/carrier bridge; the concrete two-collar geometry therefore reaches the nonempty projected theorem-4.9 endpoint itself. Lean now also exposes this nonvacuous construction at the raw quotient/error endpoint: `counterEmbedding_collarGeometry_explicitTait_nonemptyCarrier_to_theorem49RawQuotientErrorPackage` packages the collar geometry, explicit Tait coloring, nonempty purified carrier, nonempty projected synthesis, and the raw Definition 4.8 representative plus selected-boundary-kernel error decomposition for every Kirchhoff chain on the same embedding. The same file now also proves that this benchmark is not an honest closed-walk annulus source: `counterGraph_isAcyclic` identifies the graph as a three-edge matching, `not_nonempty_closedWalkAnnulusBoundarySource_counterEmbedding` blocks source data on the fixed embedding, and `counterEmbedding_rawQuotientErrorPackage_without_closedWalkAnnulusBoundarySource` bundles that source-side obstruction with the raw quotient/error package. It now also carries the repaired-witness calibration `counterEmbedding_previousBoundaryWitness_nonemptyProjectedSynthesis_without_parentWitnessOrientation`: repaired previous-boundary witness geometry and the explicit endpoint witness coexist with the nonempty projected endpoint, while the stricter construction-side parent-witness orientation still fails on the same collar. The direct replacement endpoint is therefore non-vacuous, but it is weaker than the parent-orientation route; what remains is a general boundary-order construction of the same geometry-plus-carrier package. The carrier side has now been factored into the named endpoint predicate `HasUnblockedInteriorEndpoint`, equivalent to nonempty purified carrier by `hasUnblockedInteriorEndpoint_iff_selectedBoundaryInteriorEdgeEndpointVertices_nonempty`: nonemptiness is equivalent to an interior face-incidence edge with one endpoint untouched by the selected ambient boundary. Endpoint-support disjointness plus raw `interiorEdgeEndpointSupport` nonemptiness is now a proved sufficient criterion for this local witness;

- this has now been lifted from an algebraic bridge to the actual purified theorem-4.9 endpoint. `Theorem49Synthesis.lean` proves `theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusConstruction_of_parentWitness`, which packages the full synthesis surface directly from `PlanarBoundaryAnnulusConstruction` plus `ParentWitnessOrientation`: raw-source image identity, projected/raw finite generator representations, raw Kirchhoff representatives, and the boundary-kernel decomposition are now available together with the already proved span/separation/kernel/finrank consequences. The companion graph-level theorem `exists_theorem49BoundaryRootSynthesis_of_admitsPlanarBoundaryAnnulusConstructionParentWitnessOrientation` means the live boundary-order task is no longer "reach some theorem-4.9 algebraic bridge" but specifically "produce honest parent-witness orientation on a real annulus construction";

- there is now a checked source-level obstruction showing that honest closed-walk annulus-boundary data still does not force even the weakest peeled-interior construction shell. In `Theorem49InteriorVerticesEndpointObstruction.lean`, the explicit model `diamondWithTriangleEmbedding` already carries `PlanarBoundaryClosedWalkAnnulusBoundarySource`, but the theorem `false_of_planarBoundaryAnnulusConstruction_and_hpeelInterior_diamondWithTriangle` shows that any annulus construction with the usual peeled-interior disjointness field is impossible on that embedding: the unique interior edge must be covered by a peeled witness face, yet every face containing it also meets the selected boundary support. Consequently `closedWalkAnnulusBoundarySource_does_not_imply_interiorDualBoundaryRootAdjDistancePeelData_diamondWithTriangle`, `not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_diamondWithTriangle`, `not_nonempty_planarBoundaryAnnulusConstructionBoundarySupportFaceData_diamondWithTriangle` and `closedWalkAnnulusBoundarySource_does_not_imply_planarBoundaryAnnulusConstructionBoundarySupportFaceData_diamondWithTriangle` record that the honest boundary-order source does not even reach the peeled interior-dual shell, let alone the weakest peeled-interior construction shell; moreover the same embedding already carries the stronger cyclic ordered face-arc witness `diamondWithTriangleCyclicOrderedFaceArcEmbeddingData`, so the theorems `annulusBoundaryReachability_and_cyclicOrderedFaceArcEmbeddingData_does_not_imply_interiorDualBoundaryRootAdjDistancePeelData_diamondWithTriangle` and `annulusBoundaryReachability_and_cyclicOrderedFaceArcEmbeddingData_does_not_imply_planarBoundaryAnnulusConstructionBoundarySupportFaceData_diamondWithTriangle` show that upgrading the source to cyclic ordered face-arc data still does not start the peeled interior-dual route or even the weakest peeled construction shell on this model; the same diamond witness now survives after weakening that source shell to the bundled `PlanarBoundarySelectedBoundaryArcGeometry`. The theorems `annulusBoundaryReachability_and_selectedBoundaryArcGeometry_does_not_imply_interiorDualBoundaryRootAdjDistancePeelData_diamondWithTriangle` and `annulusBoundaryReachability_and_selectedBoundaryArcGeometry_does_not_imply_planarBoundaryAnnulusConstructionBoundarySupportFaceData_diamondWithTriangle` show that even annulus roots plus one contiguous selected-boundary arc on each chosen face-boundary run still do not start the peeled interior-dual route or its weakest construction shell on this model; the explicit two-face model `doubleStarEmbedding` now blocks the source route strictly earlier. It carries both `PlanarBoundaryAnnulusBoundaryReachabilityData` and `PlanarBoundaryCyclicOrderedFaceArcEmbeddingData`, but `not_nonempty_planarBoundaryClosedWalkAnnulusBoundarySource_doubleStarEmbedding` shows that the same embedding still admits no honest facial closed-walk annulus source. The abstract summary theorem `not_forall_nonempty_planarBoundaryClosedWalkAnnulusBoundarySource_of_nonempty_planarBoundaryAnnulusBoundaryReachabilityData_and_nonempty_planarBoundaryCyclicOrderedFaceArcEmbeddingData` therefore blocks the obvious attempt to derive honest closed walks from annulus roots plus cyclic selected-boundary face arcs;

- the same double-star witness also carries the weaker bundled shell `PlanarBoundarySelectedBoundaryArcGeometry`, and the theorems `exists_embedding_withPlanarBoundaryAnnulusBoundaryReachabilityData_andPlanarBoundarySelectedBoundaryArcGeometry_withoutClosedWalkAnnulusBoundarySource` and `not_forall_nonempty_planarBoundaryClosedWalkAnnulusBoundarySource_of_nonempty_planarBoundaryAnnulusBoundaryReachabilityData_and_nonempty_planarBoundarySelectedBoundaryArcGeometry` show that even annulus roots plus one contiguous selected-boundary arc on each ambient face still do not recover the honest closed-walk annulus source;

- the positive lowering is now explicit in the source file too: `admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_cyclicOrderedFaceArcEmbeddingData_of_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource` and `admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_selectedBoundaryArcGeometry_of_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource` show that the honest closed-walk annulus source really does imply those weaker annulus-rooted source shells on the same embedding, so the double-star counterexamples are precise failed converses rather than shell mismatches;

- this converse failure is now graph-level too, not only same-embedding: `doubleStarGraph_isAcyclic` and `not_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource_doubleStarGraph` show that the underlying graph itself cannot support the honest closed-walk annulus source on any embedding, while still admitting the paired shells 'annulus reachability + cyclic ordered face arcs' and 'annulus reachability + selected-boundary arc geometry'; accordingly the new theorems `not_forall_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource_of_admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_cyclicOrderedFaceArcEmbeddingData` and `not_forall_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource_of_admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_selectedBoundaryArcGeometry` record those failed converses directly at graph level;

- the same graph-level double-star witness now blocks the natural rotation-side repairs too. The file proves `not_admitsPlanarBoundaryDartCycleEmbeddingData_doubleStarGraph`, `not_admitsPlanarBoundaryPureDartCycleEmbeddingData_doubleStarGraph`, and `not_admitsPlanarBoundaryDartSuccessorCycleEmbeddingData_doubleStarGraph`, then packages failed converses such as `not_forall_admitsPlanarBoundaryDartCycleEmbeddingData_of_` `admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_selectedBoundaryArcGeometry`, `not_forall_admitsPlanarBoundaryPureDartCycleEmbeddingData_of_admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_selectedBoundaryArcGeometry`, and `not_forall_admitsPlanarBoundaryDartSuccessorCycleEmbedding of_admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_selectedBoundaryArcGeometry`. So replacing the honest closed-walk annulus source by a dart-cycle, pure dart-cycle, or successor-cycle source does not repair the paired annulus-reachability plus selected-arc shell;

- this graph-level statement is now abstracted at the source interface itself: `PlanarBoundaryClosedWalkSource.lean` proves generic acyclic positive-edge separation theorems for the paired shells 'annulus reachability + cyclic ordered face arcs' and 'annulus reachability + selected-boundary arc geometry', plus existential failed-converse criteria; the double-star results are now concrete instances of that reusable source lemma rather than the only place where the comparison is formalized;

- the shared walk layer now abstracts the acyclic source obstruction itself. The theorem `not_nonempty_faceBoundaryClosedWalkGeometry_of_isAcyclic_of_nonempty_edgeSet` in `WalkPlanarEmbeddingWithBoundaryAcyclic.lean` shows that any acyclic graph with at least one edge cannot carry honest facial closed-walk geometry on any boundary-aware embedding, and the FourColor wrapper `not_nonempty_planarBoundaryClosedWalkEmbeddingData_of_isAcyclic_of_nonempty_edgeSet` exposes the same fact on the route-facing source interface. So the path, star, and peeled-endpoint-touch closed-walk impossibility lemmas are now instances of one reusable theorem rather than repeated local proofs; the same abstraction now reaches the stronger source shells too, via `not_admitsPlanarBoundaryDartCycleEmbeddingData_of_isAcyclic_of_nonempty_edgeSet`, `not_admitsPlanarBoundaryPureDartCycleEmbeddingData_of_isAcyclic_of_nonempty_edgeSet`, `not_admitsPlanarBoundaryDartSuccessorCycleEmbeddingData_of_isAcyclic_of_nonempty_edgeSet`, and `not_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource_of_isAcyclic_of_nonempty_edgeSet`, so acyclic positive-edge models now refute the rotation-side and explicit annulus-source interfaces without new witness-specific proofs; the same witness pattern is now packaged at the universal failed-converse level by `not_forall_admitsPlanarBoundaryClosedWalkEmbeddingData_of_exists_isAcyclic_of_nonempty_edgeSet_and_property` and `not_forall_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource_of_exists_isAcyclic_of_nonempty_edgeSet_and_property`, so the star/path failures for cyclic ordered arcs, selected-boundary arc geometry, cyclic face-boundary vertex walks, and ordered face arcs are now instances of one generic acyclic positive-edge witness criterion rather than separate local contradiction proofs; the interior-dual forcing-edge mechanism is now abstracted as `false_of_interiorDualBoundaryRootAdjDistancePeelData_of_forcing_interiorEdge`, and the weaker construction-shell version is abstracted as `false_of_planarBoundaryAnnulusConstruction_and_hpeelInterior_of_forcing_interiorEdge`, while the positive construction invariant `PlanarBoundaryAnnulusConstruction.exists_peelFace_witnessEdge_eq_and_faceBoundary_disjoint_selectedBoundarySupport_of_mem_interiorEdgeSupport` now states exactly what the weak boundary-support construction shell supplies: each interior edge is witnessed on some peeled face whose whole boundary avoids selected-boundary support. The route-facing obstruction file now packages this as the positive predicate `EveryInteriorEdgeHasBoundaryFreeIncidentFace` and proves not_nonempty_forcingInteriorEdgeWitness_iff_everyInteriorEdgeHasBoundaryFreeIncidentFace, so "no forcing interior edge" is an explicit local face-incidence obligation rather than only a failed nonemptiness conclusion. The theorem everyInteriorEdgeHasBoundaryFreeIncidentFace_of_interiorDualBoundaryRootAdjDistancePeelData now shows that the interior-dual boundary-root adjacency-distance package already supplies this local obligation directly from its peeled-edge cover and peeled-interior disjointness fields. Consequently not_nonempty_forcingInteriorEdgeWitness_of_interiorDualBoundaryRootAdjDistancePeelData is the source-side test for any proposed derivation of that package: a forcing interior edge must be impossible before the route reaches the interior-dual data surface. The theorem everyInteriorEdgeHasBoundaryFreeIncidentFace_of_planarBoundaryAnnulusConstructionBoundarySupportFaceData then records that the weak boundary-support construction shell gives exactly that obligation. Its selector form BoundaryFreeIncidentFaceSelector is now also formalized, with nonempty_boundaryFreeIncidentFaceSelector_iff_everyInteriorEdgeHasBoundaryFreeIncidentFace showing that selector nonemptiness is exactly the same local no-forcing condition. The construction shell yields the selector through boundaryFreeIncidentFaceSelector_of_planarBoundaryAnnulusConstructionBoundarySupportFaceData. The same duality is now named without the intermediate predicate: nonempty_boundaryFreeIncidentFaceSelector_iff_not_nonempty_forcingInteriorEdgeWitness and not_nonempty_boundaryFreeIncidentFaceSelector_iff_nonempty_forcingInteriorEdgeWitness identify boundary-free selector existence exactly with absence of a forcing interior edge. This is useful downstream because annulus-construction code can consume a function returning a selected-boundary-free incident face for each interior edge; it still leaves the harder compatibility requirements between those face choices for later work. The first compatibility bridge is now in Theorem49BoundaryFreeSelectorConstruction.lean. It forms the selected peel-face set BoundaryFreeIncidentFaceSelector.peelFaces, inverts the selector on that finite image through BoundaryFreeIncidentFaceSelector.selectedWitnessEdge, and proves that an injective selector with a strict-distance remainder condition lowers to PlanarBoundaryAnnulusConstruction by BoundaryFreeIncidentFaceSelector.toPlanarBoundaryAnnulusConstruction. The lemma BoundaryFreeIncidentFaceSelector.toPlanarBoundaryAnnulusConstruction_hpeelInterior then recovers the selected-boundary-free peeled-face invariant from the selector itself. The strict-descent part of this bridge now has a finite-cycle diagnostic on the selector surface: BoundaryFreeIncidentFaceSelector.remainderDependency names the relation where the selected witness of one peeled face appears as a non-witness, non-boundary remainder edge on another, BoundaryFreeIncidentFaceSelector.faceDistance_lt_of_remainderDependency turns each dependency into a strict face-distance inequality, and BoundaryFreeIncidentFaceSelector.false_of_remainderDependency_cycle rules out nonempty finite closed chains of those dependencies. The comparison theorem BoundaryFreeIncidentFaceSelector.toPlanarBoundaryAnnulusConstruction_remainderDependency_iff now identifies this selector relation with the construction-level PlanarBoundaryAnnulusConstruction.remainderDependency after lowering a strict-descent selector to PlanarBoundaryAnnulusConstruction; the construction's strict face-distance field follows from the selector hrest hypothesis. Consequently, selector-side cycle diagnostics are not merely analogous to the construction diagnostics: they transfer across the lowering bridge exactly. The no-dependency terminal condition transfers as well by `BoundaryFreeIncidentFaceSelector.toPlanarBoundaryAnnulusConstruction_no_remainderDependency_iff`. The selector-side terminal condition now has its own iff, `BoundaryFreeIncidentFaceSelector.no_remainderDependency_iff_boundary_remainders`: for a selected face, no outgoing selector dependency is exactly the statement that every non-witness remainder edge lies on the outer-or-inner ambient boundary. The maximum-distance strengthening `BoundaryFreeIncidentFaceSelector.false_of_total_remainderDependency` rules out a nonempty finite selected-face set where every selected face has an outgoing non-boundary remainder dependency, while `BoundaryFreeIncidentFaceSelector.exists_face_without_remainderDependency` extracts a terminal selected face with no such outgoing dependency. Thus a source-to-selector construction must exhibit an actual ambient-boundary break at that terminal face or add independent well-founded orientation data. The boundary-break theorem `BoundaryFreeIncidentFaceSelector.exists_terminal_face_boundary_remainders` makes this constructive: for one selected face, every non-witness remainder edge is already on the outer-or-inner ambient annulus boundary. The live-edge variants `BoundaryFreeIncidentFaceSelector.peelFaces_nonempty_of_mem_interiorEdgeSupport`, `BoundaryFreeIncidentFaceSelector.exists_terminal_face_boundary_remainders_of_mem_interiorEdgeSupport`, and `BoundaryFreeIncidentFaceSelector.exists_terminal_face_boundary_remainders_of_interiorEdgeSupport_nonempty` remove a separate selected-face nonemptiness assumption whenever an interior edge remains to be covered. The construction core now also records the non-selector version, `PlanarBoundaryAnnulusConstruction.exists_terminal_peelFace_boundary_remainders`: any nonempty BFS-stratified annulus construction has a maximum-distance peeled face whose non-witness remainders are all on the ambient outer or inner boundary. This now factors through the central strict dependency relation `PlanarBoundaryAnnulusConstruction.remainderDependency`: non-boundary remainders produce dependencies via `PlanarBoundaryAnnulusConstruction.exists_remainderDependency_of_nonboundary_remainder`, finite closed dependency chains are refuted by `PlanarBoundaryAnnulusConstruction.false_of_remainderDependency_cycle`, and the total-outgoing obstruction `PlanarBoundaryAnnulusConstruction.false_of_total_remainderDependency` exposes a peeled face with no outgoing dependency through `PlanarBoundaryAnnulusConstruction.exists_terminal_peelFace_no_remainderDependency`. The no-dependency condition has also been tied back to the local boundary break: `PlanarBoundaryAnnulusConstruction.no_remainderDependency_iff_boundary_remainders` says that a peeled face has no outgoing construction dependency exactly when all of its non-witness remainders lie on the outer-or-inner ambient annulus boundary, with the forward direction named as `PlanarBoundaryAnnulusConstruction.boundary_remainders_of_no_remainderDependency`. The core construction file now proves `PlanarBoundaryAnnulusConstruction.peelFaces_nonempty_of_mem_interiorEdgeSupport` and `PlanarBoundaryAnnulusConstruction.exists_terminal_peelFace_boundary_remainders_of_mem_interiorEdgeSupport`, so a live interior edge covered by the construction is enough to trigger the terminal boundary break. This turns the boundary break into a direct invariant of the construction's strict-descent field, without carrying peeled-face nonemptiness as a separate side condition. The orientation-obstruction file now proves the matching guardrail `not_forall_live_planarBoundaryAnnulusConstruction_terminalNoDependency_implies_parentWitnessOrientation`: even live interior-edge support plus a terminal peeled face with no outgoing construction-level `remainderDependency` does not force `PlanarBoundaryAnnulusConstruction.ParentWitnessOrientation`. Thus the terminal dependency break diagnoses unbroken strict remainder cycles, while the well-founded parent-witness orientation remains an independent geometric burden. The same concrete counter-construction now lowers to `PlanarBoundaryAnnulusConstructionPositiveFrontierData` and `PlanarBoundaryAnnulusConstructionBoundarySupportFaceData` while still refuting parent orientation, through `not_forall_planarBoundaryAnnulusConstructionPositiveFrontierData_implies_parentWitnessOrientation` and `not_forall_planarBoundaryAnnulusConstructionBoundarySupportFac` `implies_parentWitnessOrientation`. So selected-boundary contact and positive current-frontier bookkeeping are not enough to supply the missing well-founded parent witness. The raw construction obstruction is sharper: `not_forall_planarBoundaryAnnulusConstruction_positive_faceDistance_implies_parentWitnessOrientation` uses a variant where every peeled face has positive stored distance but the bad witness still has no strictly earlier incident face. Conversely, positive stored distance is not a possible repair to the positive-frontier shell itself: `PlanarBoundaryAnnulusConstructionPositiveFrontierData.exists_peelFace_faceDistance_zero` and `PlanarBoundaryAnnulusConstructionPositiveFrontierData.not_forall_peelFaces_positive` show that positive-frontier data necessarily includes a zero-distance peeled stratum. The construction core now factors this as a stratum-level clash: `PlanarBoundaryAnnulusConstruction.not_forall_stratumFaces_nonempty_of_parentWitness` shows that parent orientation makes the zero stratum empty, while the current positive-frontier axioms inhabit every stratum. The core construction file now records the stronger incompatibility as `PlanarBoundaryAnnulusConstructionPositiveFrontierData.not_parentWitnessOrientation`, with corresponding face-partition and boundary-support variants. The missing parent-orientation datum is therefore lower-distance incidence, not scalar positivity or the current zero-based positive-frontier shell. The same file now adds `not_exists_planarBoundaryAnnulusConstructionPositiveFrontierData_with_rootDistanceConstruction_of_boundaryReachabilityData_and_interiorDualBoundaryRootAdjDistancePeelData`, which makes the source-side mismatch explicit: the parent-oriented root-distance construction produced from boundary reachability plus interior-dual boundary-root adjacency-distance data cannot be wrapped in the current positive-frontier shell. Thus the root-distance parent-orientation route and the zero-based frontier bookkeeping route are different construction surfaces. The same construction file now records the exact reindexing: `rootDistance_faceDistance_succ_filter_eq_shifted_faceDistance_filter_of_boundaryReachabilityData_and_interiorDualBoundaryRootAdjDistancePeelData` says that root-distance layer $n+1$ on peeled faces is exactly shifted construction layer n, with `rootDistance_faceDistance_one_filter_eq_shifted_faceDistance_zero_filter_of_boundaryReachabilityData_and_interiorDualBoundaryRootAdjDistancePeelData` as the root-distance layer 1 / shifted layer 0 specialization. `rootDistance_faceDistance_one_filter_nonempty_iff_not_shifted_parentWitnessOrientation_of_boundaryReachabilityData_and_interiorDualBoundaryRootAdjDistancePeelData` then identifies nonempty root-distance layer 1 exactly with failure of parent-witness orientation for the shifted construction. Separately, the selector-construction file adds `BoundaryFreeIncidentFaceSelector.toPlanarBoundaryAnnulusConstructionBoundarySupportFaceData`, which reaches the stronger boundary-support face-data shell when the remaining annulus-side obligations are supplied explicitly: non-peeled faces meet selected-boundary support, outer and inner boundary faces are separated, and the positive current outer frontiers are nonempty and pairwise disjoint. The companion theorem `BoundaryFreeIncidentFaceSelector.not_nonempty_forcingInteriorEdgeWitness_of_boundarySupportFaceData_conditions` then combines that stronger bridge with the forcing-edge obstruction. Thus the unresolved burden is no longer an unspecified witness-choice problem; it is the explicit face/frontier geometry needed by the annulus construction shell. The new positive selector endpoint in `Theorem49BoundaryFreeSelectorPositiveRoute.lean` makes the useful half of this lane explicit: `BoundaryFreeIncidentFaceSelector.theorem49HeightOrderedPositiveProjectedGeometryOn_of_strictDescent` turns a boundary-free selector with annulus boundary data, injectivity over interior edges, strict-descent remainders, and a nonempty purified carrier into `Theorem49HeightOrderedPositiveProjectedGeometryOn`, while `BoundaryFreeIncidentFaceSelector.theorem49BoundaryRootNonemptyProjectedSynthesis_of_strictDescent` adds a Tait coloring and reaches the nonempty projected synthesis endpoint. It therefore specifies a concrete positive target for a boundary-order source argument without claiming that the source data already supplies the selector, strict descent, or frontier side conditions. The selector endpoint now factors through the reusable construction-level bridge `theorem49HeightOrderedPositiveProjectedGeometryOn_of_planarBoundaryAnnulusConstruction` in `Theorem49PositiveGeometricSourceReplacement.lean`: any `PlanarBoundaryAnnulusConstruction`, together with a live purified carrier on the same embedding, already supplies the height-ordered replacement package, and `theorem49BoundaryRootNonemptyProjectedSynthesis_of_planarBoundaryAnnulusConstruction` adds a Tait coloring to reach the corrected nonempty projected synthesis endpoint. Thus the selector route is one way to inhabit the annulus construction endpoint, not a separate positive surface. The same construction and live carrier hypotheses also yield `theorem49TerminalPeelFaceBoundaryRemainders_of_planarBoundaryAnnulusConstruction`, which gives a terminal peeled face whose non-witness remainders are already ambient-boundary edges. The same construction bridge now has local `HasUnblockedInteriorEndpoint` entry points: `theorem49HeightOrderedPositiveProjectedGeometryOn_of_planarBoundaryAnnulusConstruction_and_hasUnblockedInteriorEndpoint`, `theorem49BoundaryRootNonemptyProjectedSynthesis_of_planarBoundaryAnnulusConstruction_and_hasUnblockedInteriorEndpoint`, and `theorem49TerminalPeelFaceBoundaryRemainders_of_planarBoundaryAnnulusConstruction_and_hasUnblockedInteriorEndpoint`. These keep the construction target in the same local-endpoint language used by the source obstructions, instead of forcing a separate carrier-nonemptiness repackaging step. The selector positive route now has the matching local forms: `BoundaryFreeIncidentFaceSelector.theorem49HeightOrderedPositiveProjectedGeometryOn_of_strictDescent_and_hasUnblockedInteriorEndpoint`, `BoundaryFreeIncidentFaceSelector.theorem49BoundaryRootNonemptyProjectedSynthesis_of_strictDescent_and_hasUnblockedInteriorEndpoint`, and `BoundaryFreeIncidentFaceSelector.exists_terminal_face_boundary_remainders_of_strictDescent_and_hasUnblockedInteriorEndpoint`. The auxiliary `interiorEdgeSupport_nonempty_of_hasUnblockedInteriorEndpoint` is the only bridge needed to feed terminal selected-face diagnostics from a named local endpoint witness. The strengthened terminal diagnostic `BoundaryFreeIncidentFaceSelector.exists_terminal_face_boundary_remainders_and_disjoint_of_strictDescent_and_hasUnblockedInteriorEndpoint` also preserves the selector's full selected-boundary-support avoidance on that same terminal face. Thus the strict-descent endpoint now exposes a single selected face carrying both the ambient-boundary remainder break and the boundary-free incident-face condition. The wheel-with-inner-triangle benchmark now calibrates that selector burden at the source level too. The theorem not_nonempty_boundaryFreeIncidentFaceSelector_wheelWithInnerTriangle shows that this honest-source, Tait-colorable, nonempty-carrier model admits no boundary-free incident-face selector by the direct selector/forcing duality, while wheelWithInnerTriangle_closedWalkSource_tait_hasUnblockedEndpoint_without_interiorDualBoundaryRootAdjDistancePeelData states the same failure at the raw source-IDBRAD bridge boundary: honest source data, the explicit Tait coloring, and 'HasUnblockedInteriorEndpoint' coexist on the fixed embedding, but generic InteriorDualBoundaryRootAdjDistancePeelData is impossible. The theorem not_forall_some_boundaryFreeSelector_or_acyclicEndpoint_of_closedWalkSource_tait_carrier_wheel refutes any universal construction from honest closed-walk source data, a Tait coloring, and a nonempty purified carrier to a boundary-free selector or to any current acyclic annulus endpoint. The same abstract witness now also reaches the weaker paired source shell 'annulus reachability + selected-boundary arc geometry' via annulusBoundaryReachability_and_selectedBoundaryArcGeometry_does_not_imply_interiorDualBoundaryRootAdjDistancePeelData and annulusBoundaryReachability_and_selectedBoundaryArcGeometry_does_not_imply_planarBoundaryAnnulusConstructionBoundarySupportFaceData, and also the reduced construction-side theorem annulusBoundaryReachability_and_selectedBoundaryArcGeometry_does_not_imply_planarBoundaryAnnulusConstructionPositiveFronti so this obstruction applies to any embedding with an interior edge that is forced to meet selected-boundary support on every incident face, even after weakening from cyclic ordered face arcs to the bundled selected-boundary arc shell. The same forcing-edge witness now has a universal failed-converse interface: not_forall_nonempty_planarBoundaryAnnulusConstructionBoundarySupportFaceData_of_exists_embedding_with_property_and_forcingInteriorEdgeWitness and not_forall_nonempty_planarBoundaryAnnulusConstructionBoundarySupportFaceData_of_exists_embedding_closedWalkAnnulusBoundarySource_and_forcingInteriorEdgeWitness show that any source property coexisting with such a forcing edge cannot by itself derive the weakest peeled-interior construction shell; the regression file now also phrases this at graph level in same-embedding existential form on diamondWithTriangleGraph: there is an embedding with the weaker source shells plus a forcing witness, but no embedding of that same graph can combine the witness with the peeled interior-dual package, the reduced positive-frontier shell, or the concrete PlanarBoundaryAnnulusConstructionFaceLayerData shell that lowers into annulus decomposition; the same forcing witness now also blocks the current two-sided annulus-root lane itself, via closedWalkAnnulusBoundarySource_does_not_imply_planarBoundaryAnnulusRootAdjDistancePeelData and not_exists_embedding_closedWalkAnnulusBoundarySource_and_forcingInteriorEdgeWitness_and_planarBoundaryAnnulusRootAdjDistancePeelData, so boundary-split repackaging does not rescue a forcing-edge source embedding; this now reaches one step earlier too, since the same forcing witness blocks PlanarBoundaryAnnulusRootAdjDistancePeelData already from PlanarBoundaryAnnulusBoundaryReachabilityData+PlanarBoundaryCyclicOrderedFaceArcEmbeddingD and from the weaker bundled PlanarBoundarySelectedBoundaryArcGeometry shell; and this is now packaged as a reusable failed-converse surface by not_forall_nonempty_planarBoundaryAnnulusRootAdjDistancePeelData_of_exists_embedding_with_property_and_forcingInteriorEdgeWitness and its three route-facing specializations, so the forcing-edge blocker is not merely a family of diamond-with-triangle regressions but a same-embedding "no universal converse" theorem surface for the current annulus-root lane; more sharply, the same file now also proves `not_forall_not_nonempty_forcingInteriorEdgeWitness_of_exists_embedding_with_property_and_forcingInteriorEdgeWitness` and its route-facing specializations, so the current source-side shells do not even universally exclude forcing interior edges on one embedding; and `PlanarBoundaryClosedWalkSource.lean` now makes explicit that an honest closed-walk source already lowers on the same embedding to the cyclic ordered face-arc and bundled selected-boundary-arc shells, while `Theorem49ForcingInteriorEdgeObstruction.lean` shows that those stronger honest-source shells still do not imply annulus-root data in the presence of a forcing interior edge; the same file now also records the graph-level same-embedding obstructions for those strengthened honest-source conjunctions, so "closed-walk source + cyclic face-arc shell" and "closed-walk source + bundled selected-boundary arc shell" are both formally blocked from coexisting with annulus-root data when a forcing witness is present; `Theorem49ForcingInteriorEdgeObstruction.lean` now also records that the natural face-local repair is insufficient. The proved failed-converse theorem `not_forall_not_nonempty_forcingInteriorEdgeWitness_of_exists_embedding_reachability_and_closedWalkEmbedding_and_atMostOneInteriorEdge_and_forcingInteriorEdgeWitness` is instantiated in `Theorem49ForcingInteriorEdgeObstructionRegression.lean` by `exists_embedding_reachability_and_closedWalkEmbedding_and_atMostOneInteriorEdge_and_forcingInteriorEdgeWitness_diamondWithTriangleGraph` and `not_forall_not_nonempty_forcingInteriorEdgeWitness_of_reachability_and_closedWalkEmbedding_and_atMostOneInteriorEdge_diamondWithTriangle`: on `diamondWithTriangleEmbedding`, annulus boundary reachability, honest facial closed walks, and at most one interior boundary edge on each face coexist with the forcing interior edge `dt12`. So the forcing-edge exclusion burden cannot be discharged merely by adding an at-most-one interior-edge condition to the closed-walk source surface. The same regression file now also records the direct route implication of this model through `diamondWithTriangle_source_previousBoundaryWitness_and_witnessCover_without_interiorDual_rootDistance`: the embedding has honest closed-walk annulus source data, repaired previous-boundary witness geometry, and both well-founded and height-ordered witness-cover data, while the forcing edge blocks both `InteriorDualBoundaryRootAdjDistancePeelData` and the two-sided `PlanarBoundaryAnnulusRootAdjDistancePeelData`. The failed-converse theorems `not_forall_interiorDualBoundaryRootAdjDistancePeelData_of_previousBoundaryWitnessGeometry_diamondWithTriangle` and `not_forall_planarBoundaryAnnulusRootAdjDistancePeelData_of_closedWalkAnnulusBoundarySource_and_previousBoundaryWitnessGeometry_diamondWithTriangle` make the route consequence explicit: root-distance interior-dual data cannot be derived from the repaired previous-boundary witness surface, even with the honest source retained on the same graph;

- the later peeled endpoint-touch model is now calibrated against the source side too. Although `peeledEndpointTouchEmbedding` carries the current outer-boundary-rooted peeled annulus package, the graph `peeledEndpointTouchGraph` is acyclic, so `not_nonempty_planarBoundaryClosedWalkEmbeddingData_peeledEndpointTouchGraph` and `not_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph` show that it cannot carry the honest facial closed-walk source at all. Consequently `admitsPlanarBoundaryOuterBoundaryRootAdjDistancePeelData_withoutClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph` and `admitsPlanarBoundaryHeightOrderedFacePeelWitnessData_withoutClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph` and `admitsPlanarBoundaryWellFoundedFacePeelWitnessData_withoutClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph` and the new `admitsPlanarBoundaryAnnulusRootAdjDistancePeelData_withoutClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph` and `admitsPlanarBoundaryAttachedRootedFacePeelCertificate_withoutClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph` should be read as calibration theorems about later route layers, not as source-level counterexamples;

- the endpoint-separation package has now reached a natural positive completion point. In `Theorem49InteriorVertices.lean`, the theorem `PlanarBoundaryOuterBoundaryRootAdjDistancePeelDataWithEndpointSeparation.selectedBoundaryInteriorEdgeEndpointVertices_eq_interiorEdgeEndpointSupport` shows that the packaged endpoint-disjointness field makes selected-boundary purification lossless on the raw interior-edge endpoint carrier. Then `Theorem49Synthesis.lean` proves `theorem49OuterBoundaryRootNonemptySynthesis_of_planarBoundaryOuterBoundaryRootAdjDistancePeelDataWithEndpointSeparation`: endpoint-separated outer-boundary-rooted annulus data plus nonempty raw `interiorEdgeEndpointSupport` already yields the non-vacuous Theorem 4.9 synthesis package, with graph-level wrapper `exists_theorem49OuterBoundaryRootNonemptySynthesis_of_exists_planarBoundaryOuterBoundaryRootAdjDistancePeelDataWithEndpointSeparation_with_nonempty_interiorEdgeEndpointSupport`. The route now also has the matching construction/source bridge: `theorem49OuterBoundaryRootNonemptySynthesis_of_planarBoundaryOuterBoundaryRootAdjDistancePeelData_of_peelFaces_endpoint_disjoint_selectedBoundarySupport` lowers peeled-face endpoint no-touch directly to that non-vacuous endpoint, and `PlanarBoundaryClosedWalkSourceRoute.lean` packages the honest closed-walk source analogue under explicit outer-root localization and raw carrier nonemptiness. However, the same file now also proves `not_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_and_rootsOuterAmbientBoundary_and_peelFaces_endpoint_disjoint_selectedBoundarySupport_and_nonempty_interiorEdgeEndpointSupport` and the endpoint-level form `not_exists_closedWalkAnnulusBoundarySource_and_theorem49OuterBoundaryRootNonemptySynthesis_sameBoundaryData`. So this outer-root repair line is not a same-split rescue of the honest closed-walk route: once the annulus boundary split is fixed to the source's own extracted split, the needed outer-root localization witness is already inconsistent. The replacement positive line is now the two-sided annulus-root surface instead. `Theorem49Synthesis.lean` proves `theorem49AnnulusRootNonemptySynthesis_of_planarBoundaryAnnulusRootAdjDistancePeelData_of_endpointSupport_disjoint` and `exists_theorem49AnnulusRootNonemptySynthesis_of_exists_planarBoundaryAnnulusRootAdjDistancePeelData_with_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport`, and the peeled-face repair has now been lifted to the same generic surface. `Theorem49InteriorVerticesAnnulusConstruction.lean` defines `planarBoundaryAnnulusConstruction_of_planarBoundaryAnnulusRootAdjDistancePeelData` and proves the annulus-root lossless-purification bridge `PlanarBoundaryAnnulusRootAdjDistancePeelData.selectedBoundaryInteriorEdgeEndpointVertices_eq_interiorEdgeEndpointSupport_of_peelFaces_endpoint_disjoint_selectedBoundarySupport`. `Theorem49Synthesis.lean` then proves `theorem49AnnulusRootNonemptySynthesis_of_planarBoundaryAnnulusRootAdjDistancePeelData_of_peelFaces_endpoint_disjoint_selectedBoundarySupport` and the graph-level wrapper `exists_theorem49AnnulusRootNonemptySynthesis_of_exists_planarBoundaryAnnulusRootAdjDistancePeelData_with_peelFaces_endpoint_disjoint_selectedBoundarySupport_and_nonempty_interiorEdgeEndpointSupport`. `PlanarBoundaryClosedWalkSourceRoute.lean` packages the honest closed-walk source specializations `theorem49AnnulusRootNonemptySynthesis_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_of_endpointSupport_disjoint` and `exists_theorem49AnnulusRootNonemptySynthesis_of_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_with_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport`. It also keeps the source-facing peeled-face forms `theorem49AnnulusRootNonemptySynthesis_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_of_peelFaces_endpoint_disjoint_selectedBoundarySupport` and `exists_theorem49AnnulusRootNonemptySynthesis_of_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_with_peelFaces_endpoint_disjoint_selectedBoundarySupport_and_nonempty_interiorEdgeEndpointSupport` as specializations of the generic bridge, deriving global raw-carrier endpoint disjointness from `PlanarBoundaryAnnulusConstruction.endpointSupport_disjoint_of_peelFaces_endpoint_disjoint_selectedBoundarySupport`. So the current non-vacuous repair no longer depends on forcing every root to the extracted outer boundary, and the peeled-face no-touch route is available for any explicit two-sided annulus-root datum, not only data produced from an honest closed-walk source. This extra hypothesis is not automatic from current annulus-root data alone: `Theorem49InteriorVerticesEndpointObstruction.lean` now proves `annulusRootAdjDistancePeelData_does_not_imply_endpointSupport_disjoint_peeledEndpointTouch` and the fixed-embedding universal failure `not_forall_annulusRootAdjDistancePeelData_implies_endpointSupport_disjoint_peeledEndpointTouch`. So the new annulus-root repair theorem is formally honest about its missing geometric burden instead of hiding it inside the present route data. The honest annulus-geometry lane now calibrates the same point from the source side: `Theorem49PlanarBoundaryAnnulusHonestGeometryRegression.lean` proves `closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_and_attachedRootedFacePeelCertificate_with_nonempty_interiorEdgeEndpointSupport_without_endpointSupport_disjoint_diamondWithTriangle`, so on `diamondWithTriangleEmbedding` an honest closed-walk annulus source, weak one-collar annulus geometry, the attached certificate endpoint, and nonempty raw `interiorEdgeEndpointSupport` already coexist while endpoint-support disjointness still fails. Thus the current endpoint- separation repair sits strictly later than the honest collar/certificate lane too. The stronger repaired geometry surface also fails to force it: `closedWalkAnnulusBoundarySource_and_previousBoundaryWitnessGeometry_and_wellFoundedFacePeelWitnessData_with_nonempty_interiorEdgeEndpointSupport_without_endpointSupport_disjoint_diamondWithTriangle` shows that even previous-boundary witness geometry and the derived boundary-aware well-founded peel data can coexist with a nonempty raw carrier while endpoint-support disjointness still fails on the same honest source model. And the stronger theorem `closedWalkAnnulusBoundarySource_and_previousBoundaryWitnessGeometry_and_wellFoundedFacePeelWitnessData_with_nonempty_interiorEdgeEndpointSupport_and_empty_selectedBoundaryInteriorEdgeEndpointVertices_diamondWithTriangle` now shows that this is not just a failure of equality between the raw and purified carriers: on the same honest source model the raw `interiorEdgeEndpointSupport` is nonempty while the purified `selectedBoundaryInteriorEdgeEndpointVertices` is exactly empty. The same file now pushes this one layer further: the explicit collar-layer data `diamondWithTriangleOneCollarLayerFacePeelWitnessData` and its induced height-ordered peel witness `diamondWithTriangleOneCollarHeightOrderedFacePeelWitnessData` already exist on the same honest source model, yet `closedWalkAnnulusBoundarySource_and_previousBoundaryWitnessGeometry_and_heightOrderedFacePeelWitnessData_with_nonempty_interiorEdgeEndpointSupport_and_empty_selectedBoundaryInteriorEdgeEndpointVertices_diamondWithTriangle` shows that the purified carrier is still empty there. So the endpoint-touch obstruction and the endpoint-separation repair line now meet cleanly on the same theorem surface;

- the theorem-4.9 synthesis layer is now generic above the boundary-aware certificate surface. `Theorem49Synthesis.lean` proves `theorem49BoundaryRootSynthesis_of_boundaryZeroAnnihilatorTrivial`, then packages the full purified endpoint directly from `PlanarBoundaryAttachedRootedFacePeelCertificate` via `theorem49BoundaryRootSynthesis_of_planarBoundaryAttachedRootedFacePeelCertificate` and from `PlanarBoundaryWellFoundedFacePeelWitnessData` via `theorem49BoundaryRootSynthesis_of_planarBoundaryWellFoundedFacePeelWitnessData`, with matching graph-level wrappers. The older interior-dual and annulus-construction synthesis theorems now reduce through the same generic bridge, so once an honest route reaches the attached certificate or well-founded peel witness surface, no further theorem-4.9 algebra remains. More sharply, `Theorem49Synthesis.lean` now proves `theorem49BoundaryRootSynthesis_of_planarBoundaryHeightOrderedFacePeelWitnessData`, `theorem49BoundaryRootSynthesis_of_planarBoundaryCollarLayerFacePeelWitnessData`, `theorem49BoundaryRootSynthesis_of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData`, `theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusWitnessAssignment`, and the graph-level wrappers `exists_theorem49BoundaryRootSynthesis_of_admitsPlanarBoundaryHeightOrderedFacePeelWitnessData`, `exists_theorem49BoundaryRootSynthesis_of_admitsPlanarBoundaryCollarLayerFacePeelWitnessData`, `exists_theorem49BoundaryRootSynthesis_of_admitsPlanarBoundaryLocalRemainderBoundaryCollarL` and `exists_theorem49BoundaryRootSynthesis_of_admitsPlanarBoundaryAnnulusWitnessAssignment`, so theorem-4.9 synthesis already starts at the boundary-aware height-ordered witness-cover surface and even at the weaker finite collar-layer witness-cover surface, hence also at the weakest explicit local remainder annulus surface and at the split "annulus decomposition + witness assignment" surface. `admitsPlanarBoundaryAttachedRootedFacePeelCertificate_of_` admitsPlanarBoundaryAnnulusConstructionParentWitnessOrientation, so the repaired parent-orientation route already lands on the certificate surface explicitly, and the direct honest-source synthesis corollaries in PlanarBoundaryClosedWalkSourceRoute.lean now factor through that certificate surface rather than bypassing it. The same source file now also proves exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_heightOrderedFacePeelWitnessData_direct, exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_collarLayerFacePeelWitnessData_direct, exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_annulusWitnessAssignment_direct and exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_heightOrderedFacePeelWitnessData_direct, exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_collarLayerFacePeelWitnessData_direct, exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_annulusWitnessAssignment_direct, so the honest source shell already reaches theorem-4.9 synthesis as soon as the same embedding carries either boundary-aware height-sorted witness-cover data, the weaker finite collar-layer witness-cover package, or the later pure annulus decomposition together with witness-edge ownership. The same route file now also proves the matching direct Step 2 vanishing corollaries zero_on_allEdges_of_exists_closedWalkAnnulusBoundarySource_and_heightOrderedFacePeelWitnessData_direct_of_annihilatesKempeClosureGeneratorFamily, zero_on_allEdges_of_exists_closedWalkAnnulusBoundarySource_and_collarLayerFacePeelWitnessData_direct_of_annihilatesKempeClosureGeneratorFamily, zero_on_allEdges_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_heightOrderedFacePeelWitnessData_direct_of_annihilatesKempeClosureGeneratorFamily, and zero_on_allEdges_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_collarLayerFacePeelWitnessData_direct_of_annihilatesKempeClosureGeneratorFamily. So the positive algebraic boundary now matches the positive synthesis boundary: once an honest source shell reaches boundary-aware height-sorted witness-cover data, or even just the finite collar-layer witness-cover package, no later annulus-geometry packaging is needed either for Step 2 vanishing or for the purified theorem-4.9 endpoint. Theorem49Synthesis.lean also proves theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusCollarGeometry, theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusPreviousBoundaryWitnessGeometry, and the graph-level wrappers exists_theorem49BoundaryRootSynthesis_of_admitsPlanarBoundaryAnnulusCollarGeometry, exists_theorem49BoundaryRootSynthesis_of_admitsPlanarBoundaryAnnulusPreviousBoundaryWitnessGeomet and exists_theorem49BoundaryRootSynthesis_of_exists_planarBoundaryAnnulusCollarGeometry_and_witnessOnCurrentBoundary. So the synthesis burden itself no longer waits even for bundled collar geometry: once the route produces the boundary-aware height-ordered witness-cover surface, or even the weaker finite collar-layer witness-cover package, the certificate layer and the full current theorem-4.9 endpoint are automatic; the live positive burden is now upstream of explicit remainder bookkeeping and witness ownership themselves; the geometry file `Theorem49PlanarBoundaryAnnulusGeometry.lean` now makes that burden explicit by constructing the witness-cover surfaces directly from annulus geometry itself: `PlanarBoundaryAnnulusCollarGeometry.toPlanarBoundaryCollarLayerFacePeelWitnessData`, `PlanarBoundaryAnnulusCollarGeometry.toPlanarBoundaryHeightOrderedFacePeelWitnessData`, and the graph-level wrappers from admitted collar geometry, plus the repaired lane `admitsPlanarBoundaryWellFoundedFacePeelWitnessData_of_exists_planarBoundaryAnnulusCollarGeometry_and_witnessOnCurrentBoundary`. Then `PlanarBoundaryClosedWalkSourceRoute.lean` adds the matching honest closed-walk and successor-cycle wrappers from same-embedding annulus collar geometry to collar-layer witness-cover, height-ordered witness-cover, and, with current-boundary witness placement, the well-founded peel witness surface. So the remaining live route burden is cleanly geometric: derive same-embedding annulus collar geometry from honest source hypotheses, and add the repaired witness placement only if one wants the well-founded lane;

- Lean now isolates one explicit sufficient criterion for that first geometric step on the canonical one-collar route. `Theorem49PlanarBoundaryAnnulusWitness.lean` defines `PlanarBoundaryCanonicalWitnessChoiceboundaryData`, a facewise choice of one boundary edge on each ambient face of a fixed annulus boundary split such that all interior edges are chosen by some face and every non-witness boundary edge already lies on the outer or inner ambient annulus boundary. `PlanarBoundaryCanonicalWitnessChoice.toPlanarBoundaryAnnulusWitnessAssignment` then upgrades the canonical decomposition `planarBoundaryAnnulusDecomposition_of_boundaryDataboundaryData` to a full annulus witness assignment, and `PlanarBoundaryClosedWalkSourceRoute.lean` packages the honest-source corollary as `admitsPlanarBoundaryAnnulusCollarGeometry_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_direct`. So the live one-collar construction burden is now stated on a concrete facewise object rather than an opaque bundled collar-geometry shell;

- the stronger cyclic ordered face-arc source object is now equivalent, on one embedding and at graph level, to that split cyclic walk-plus-contiguity shell, so future honest embedding-side work may target either theorem surface without changing the downstream annulus route;

- on the theorem-4.9 annulus side, the bundled collar geometry object is now also exposed as an exact packaging of the split surface "pure annulus decomposition + witness assignment", both on one embedding and at graph level, so downstream work may use whichever form is more convenient without changing the annihilator route;

- the newer residual/current-boundary wrapper is now calibrated just as sharply on the positive side. The file `Theorem49ResidualBoundaryPeeling.lean` proves `nonempty_planarBoundaryResidualBoundaryLayerFacePeelWitnessData_iff_nonempty_planarBoundaryCollarLayerFacePeelWitnessData`, `theorem49ResidualBoundaryPositiveProjectedGeometryOn_iff_collarLayerPositiveProjectedGeometryOn`, and `theorem49ResidualBoundaryPositiveProjectedGeome iff_heightOrderedPositiveProjectedGeometryOn`. So at the present theorem-4.9 endpoint the residual/current-boundary surface is formally conservative: it is exactly the old collar-layer / height-ordered witness-cover strength on a fixed embedding, not yet a stronger same-embedding source construction theorem;

- the same negative calibration now reaches the exact v23 Step 2 seed itself. In `Theorem49ResidualBoundaryObstruction.lean`, `nonempty_sharedInteriorPair_v23ResidualBoundaryInitialState_sipFace0Boundary` builds `V23ResidualBoundaryInitialState` on a genuine face boundary of the shared-interior-pair source, while `sharedInteriorPair_closedWalkSource_tait_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_residualBoundaryGeometry` shows that this exact algebraic seed, together with honest source data, a Tait coloring, and the purified endpoint witness, still does not force residual positive geometry on the same embedding. So the live missing theorem is a real source-to-residual geometric bridge, not better packaging of the existing single-component purification algebra;

- the same sharp failure now holds already on the route-facing successor-cycle boundary-order shell. The theorem `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_residualBoundaryPositiveProjectedGeometryOn_sharedInteriorPair` packages explicit annulus boundary reachability, selected-boundary arcs from successor cycles, the concrete Tait coloring, ‘HasUnblockedInteriorEndpoint‘, and the exact v23 residual seed together with failure of residual positive geometry on the same embedding. Its universal converse `not_forall_residualBoundaryPositiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` shows that the missing bridge is already absent at the actual boundary-order route surface, before any lowering or repackaging to closed-walk source form;

- the exact v23 residual seed also fails to recover any previously accepted same-embedding geometry surface. The theorem `sharedInteriorPair_closedWalkSource_tait_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_knownSameEmbeddingGeometry` packages the honest source, concrete Tait coloring, ‘HasUnblockedInteriorEndpoint‘, and exact seed together with failure of residual witness data, residual selector data, height-ordered witness data, collar-layer witness data, the attached boundary-rooted certificate, well-founded witness data, and weak annulus-collar geometry on the same embedding. The failed universals `not_forall_some_knownSameEmbeddingGeometry_of_closedWalkAnnulusBoundarySource_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` and `not_forall_some_knownSameEmbeddingGeometry_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` show that this exact-seed gap already persists on both the honest closed-walk and the actual successor-cycle boundary-order shells. So the live v23.5 obligation is a genuinely new source-to-residual bridge, not a return to any existing same-embedding geometry ladder;

- the annulus witness file now exposes the actual parent-carrying geometry that had previously been hidden inside the previous-boundary witness endpoint.

PlanarBoundaryAnnulusDecomposition.layerOf gives the canonical collar index of an ambient face, and PlanarBoundaryAnnulusWitnessAssignment.exists_parentFace_of_previousBoundaryWitness proves that every positive-collar witness edge placed on the previous annulus boundary has an adjacent face in a strictly earlier collar layer. The chosen map PlanarBoundaryAnnulusWitnessAssignment.parentFaceOfPreviousBoundaryWitness is then used directly in the well-founded witness construction. So the live geometric burden is now explicit: prove that previous-boundary witness placement, or an honest stronger substitute such as annular no-touch, from real boundary-order data;

- even this repaired witness-placement surface should not be sent back through the canonical BFS construction as if it implied the construction-side ParentWitnessOrientation shell. The new regression file Theorem49PlanarBoundaryAnnulusGeometryRegression.lean gives a two-face model carrying PlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry, equivalently WitnessOnCurrentBoundary, whose canonical lowering planarBoundaryAnnulusConstruction_of_annulusCollarGeometry still fails ParentWitnessOrientation because the outer collar face itself has construction distance 0. The same repaired model already carries the attached boundary-rooted face-peel certificate reached by the direct annulus route, so the canonical construction shell is now explicitly shown to be strictly unnecessary on that witness. More sharply, the live Lean route now shows that weak collar geometry already reaches the attached certificate surface, while the repaired previous-boundary witness condition is only needed for the well-founded parent-peeling branch. So the honest ground-up annulus route is the direct geometry $\to$ certificate bridge together with the repaired-geometry $\to$ well-founded-witness bridge, not "repaired geometry $\to$ canonical construction $\to$ parent orientation";

- that route picture is now witnessed on an honest source model too. The new file Theorem49PlanarBoundaryAnnulusHonestGeometryRegression.lean equips the genuine closed-walk source model diamondWithTriangleEmbedding with a weak one-collar annulus geometry diamondWithTriangleOneCollarGeometry. On that same embedding the weak geometry already yields the attached certificate diamondWithTriangleOneCollarGeometry_hasAttachedRootedFacePeelCertificate, while the canonical lowering diamondWithTriangleOneCollarConstruction still fails ParentWitnessOrientation by not_diamondWithTriangleOneCollarConstructionParentWitnessOrientation. Even more sharply, the theorem closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_and_attachedRootedFacePeelCertificate_without_interiorDualBoundaryRootAdjDistancePeelData_diamondWithTriangle shows that the honest facial closed-walk source, weak annulus geometry, and attached certificate coexist on one embedding even though the current interior-dual boundary-root adjacency-distance package still fails there. So the direct certificate lane is not merely earlier than canonical parent orientation; it is already earlier than the present interior-dual route on a genuine cyclic source model;

- this source-side annulus lane now reaches the Step 2 annihilator endpoint directly as well. In PlanarBoundaryClosedWalkSourceRoute.lean, zero_on_allEdges_of_exists_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_direct_of_annihilatesKempeClosureGeneratorFamily and its successor-cycle analogue zero_on_allEdges_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_annulusCollarGeometry_direct_of_annihilatesKempeClosureGeneratorFamily show that honest source data plus same-embedding weak collar geometry, boundary vanishing, and the Kempe-generator annihilation hypothesis already force global vanishing, without routing through the present `InteriorDualBoundaryRootAdjDistancePeelData` package. The same file now also proves `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_direct` and `exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_annulusCollarGeometry_direct`, so once honest source data is paired with same-embedding weak collar geometry, the full current theorem-4.9 synthesis endpoint is automatic;

- under the current weak API, even pure annulus decomposition is no longer an open burden on the honest source lane. In `Theorem49PlanarBoundaryAnnulusDecomposition.lean`, `planarBoundaryAnnulusDecomposition_of_boundaryData` builds a trivial one-collar decomposition directly from the annulus boundary split, and `PlanarBoundaryClosedWalkSource.lean` packages that as `admitsPlanarBoundaryAnnulusDecomposition_of_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource`. So the source shell already reaches pure decomposition; the real remaining burden is witness-edge ownership and stronger collar geometry, not the naked existence of collars;

- `Theorem49PlanarBoundaryAnnulusWitnessRegression.lean` now shows that this is a real extra burden, not just bookkeeping. The generic theorem `not_nonempty_planarBoundaryAnnulusWitnessAssignment_of_exists_two_distinct_interior_edges_on_faceBoundary_of_canonicalDecomposition` in `Theorem49PlanarBoundaryAnnulusWitness.lean` shows that any canonical one-collar decomposition with a face carrying two distinct interior edges has no witness assignment at all, and the concrete annulus boundary model in `Theorem49PlanarBoundaryAnnulusWitnessRegression.lean` is an explicit instance. So trivializing pure decomposition from boundary data does not trivialize Step 2 witness ownership;

- that same obstruction now lives on the honest source surface too. The new theorem `not_nonempty_planarBoundaryAnnulusWitnessAssignment_of_exists_two_distinct_interior_edges_on_faceBoundary_of_closedWalkAnnulusBoundarySource` pushes the canonical one-collar blocker through an honest `PlanarBoundaryClosedWalkAnnulusBoundarySource`, and the new file `Theorem49PlanarBoundaryAnnulusHonestWitnessRegression.lean` realizes it on a finite source model where two outer faces share the consecutive interior edges `sip01` and `sip12` while a disjoint inner triangle supplies the second boundary component. That model carries the full honest source package but its canonical decomposition still has no witness assignment, so the repair "honest source ⇒ boundary data ⇒ canonical decomposition" is blocked on a genuine same-embedding source witness;

- the newer explicit canonical-witness-choice repair is blocked by the same geometry. In `Theorem49PlanarBoundaryAnnulusWitness.lean`, `not_nonempty_planarBoundaryCanonicalWitnessChoice_of_exists_two_distinct_interior_edges_on_faceBoundary` shows that a face carrying two distinct interior edges cannot support the canonical facewise witness choice at all, because such a choice would upgrade to the forbidden canonical one-collar witness assignment. The abstract boundary-data failed converse is `not_forall_planarBoundaryAnnulusBoundaryData_implies_nonempty_canonicalWitnessChoice`, and the honest source failed converse is `not_forall_exists_planarBoundaryCanonicalWitnessChoice_of_closedWalkAnnulusBoundarySource_sharedInteriorPair`. Thus the positive theorem `admitsPlanarBoundaryAnnulusCollarGeometry_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_direct` has a Lean-calibrated extra geometric hypothesis, not a hidden consequence of the honest source shell. That positive hypothesis now constructs actual one-collar geometry on the same embedding: `PlanarBoundaryCanonicalWitnessChoice.toPlanarBoundaryAnnulusCollarGeometry` turns any canonical witness choice on any annulus boundary split into a `PlanarBoundaryAnnulusCollarGeometry`, while `PlanarBoundaryCanonicalWitnessChoice.toPlanarBoundaryAnnulusCollarGeometry_numCollars` and `PlanarBoundaryCanonicalWitnessChoice.toPlanarBoundaryAnnulusCollarGeometry_boundaryData` prove that the geometry has exactly one collar and preserves the original boundary split. The honest-source theorem `exists_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice` is the broader class version: every honest closed-walk source whose boundary split carries a canonical facewise witness choice has same-embedding one-collar collar geometry with that source boundary split. Lean now also constructs the canonical witness choice from a still more local face condition: `PlanarBoundaryCanonicalWitnessChoice.of_atMostOneInteriorEdgePerFace` uses the unique interior edge on a face when one exists and a supplied fallback boundary edge otherwise, assuming every face has at most one interior boundary edge and every non-interior face-boundary edge lies on the outer or inner ambient annulus boundary. The latter assumption is now discharged by annulus boundary data itself: `PlanarBoundaryAnnulusBoundaryData.mem_outerAmbientBoundary_union_innerAmbientBoundary_of_mem_faceBoundary_of_not_mem_interiorEdgeSupport` classifies any non-interior face-boundary edge as selected boundary, hence as outer or inner ambient boundary. Lean now also gets the fallback edge directly from the honest facial closed walks: `PlaneEmbeddingWithBoundary.FaceBoundaryClosedWalk.faceBoundary_nonempty` turns each nonempty face-boundary walk into a nonempty face-boundary edge set, and `PlanarBoundaryClosedWalkAnnulusBoundarySource.fallbackEdge` chooses a boundary edge on every ambient face. Thus `exists_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_of_exists_closedWalkAnnulusBoundarySource_and_facewiseUniqueInteriorEdge` needs only the equality-style facewise uniqueness of interior boundary edges, and the route-side cardinal theorem `exists_planarBoundaryCanonicalWitnessChoice_of_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge` needs only the finite at-most-one interior-edge bound. The older explicit source wrapper `exists_planarBoundaryCanonicalWitnessChoice_of_closedWalkAnnulusBoundarySource_and_atMostOneInteriorEdgePerFace` is now complemented by the exact converse criterion in `PlanarBoundaryClosedWalkSourceCanonicalWitness.lean`: `nonempty_planarBoundaryCanonicalWitnessChoice_iff_facewiseAtMostOneInteriorEdge_of_closedWalkAnnulusBoundarySource` identifies canonical witness existence on an honest closed-walk source with the facewise at-most-one bound, and `exists_closedWalkAnnulusBoundarySource_and_planarBoundaryCanonicalWitnessChoice_iff_exists_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge` records the same equivalence at the graph-level source-search surface. The canonical-witness source attack is therefore exactly the at-most-one face obligation already isolated by the obstruction files. still records the fully expanded fallback/count/boundary-rest package. The same cardinal hypothesis now also reaches fixed-embedding collar geometry: `exists_oneCollarAnnulusCollarGeometry_of_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge` constructs one-collar `PlanarBoundaryAnnulusCollarGeometry` preserving the source boundary split from an honest source and facewise at-most-one alone. Thus the minimal extra geometric hypothesis at this layer is precisely the finite at-most-one condition; fallback and non-interior boundary-rest are source consequences. The graph-level endpoint `exists_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_of_exists_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdgeCardinality` now packages this as an existential source theorem: an honest closed-walk annulus source plus the facewise cardinal bound directly yields same-embedding one-collar collar geometry preserving the source boundary split. The route file now also repairs this branch to the previous-boundary witness surface: `admitsPlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry_of_exists_oneCollarAnnulusCollarGeometry` uses the fact that a one-collar geometry has no positive collar index, and `admitsPlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry_of_exists_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge_direct` applies that repair to the honest-source cardinal criterion without fallback or boundary-rest assumptions. Thus the local at-most-one package reaches the corrected previous-boundary, height-ordered, and collar-layer witness surfaces without another geometric placement hypothesis. The same repaired landing is now available directly from the canonical witness choice surface: `PlanarBoundaryCanonicalWitnessChoice.toPlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry` builds `PlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry` from the facewise canonical choice, and `exists_planarBoundaryAnnulusPreviousBoundaryWitnessGeometry_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice` packages the honest-source existential form. The result is specific to the canonical one-collar geometry induced by the source boundary split; arbitrary weak collar geometries still have to carry `WitnessOnCurrentBoundary` explicitly. The route theorem `admitsPlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_direct` now lands through this repaired geometry theorem directly, so the canonical source route no longer needs an intermediate one-collar existence detour at this boundary. The route file now also pushes the cardinal form of this local criterion to the theorem-4.9 synthesis endpoint: `exists_planarBoundaryCanonicalWitnessChoice_of_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge` constructs canonical witness choice from the cardinal at-most-one hypothesis alone, and `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge_direct` then produces `Theorem49BoundaryRootSynthesis` for any Tait coloring. The same source package now has an explicit nonempty-carrier endpoint in the caller-clean cardinal form: `theorem49BoundaryRootNonemptySynthesis_of_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge_direct` adds only `(selectedBoundaryInteriorEdgeEndpointVerticesemb).Nonempty`, and `exists_theorem49BoundaryRootNonemptySynthesis_of_exists_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge_with_nonempty_selectedBoundaryInteriorEdgeEndpointVertices_direct` packages the graph-level form. The positive nonempty burden is therefore a Tait-colorable honest source satisfying facewise cardinal at-most-one while retaining a nonempty purified carrier, with no caller-supplied fallback or boundary-rest proof. The same source-side package now reaches the canonical Definition 4.8 generator spanning bridge. The direct wrappers `exists_theorem49_kempeGeneratorSubspace_spanning_of_exists_closedWalkAnnulusBoundarySource_and_atMostOneInteriorEdgePerFace_heightOrdered_direct` and `exists_theorem49_kempeGeneratorSubspace_spanning_of_exists_closedWalkAnnulusBoundarySource_and_atMostOneInteriorEdgePerFace_collarLayer_direct` compose the repaired one-collar source route with the canonical `Theorem49KempeGeneratorSpan.lean` wrappers, so the only remaining spanning inputs are the target subspace, containment of the raw Kempe-generator span, and boundary-zero interpretation. The stronger one-collar geometry surface also now extracts this hypothesis directly: `PlanarBoundaryAnnulusCollarGeometry.toPlanarBoundaryCanonicalWitnessChoice_of_numCollars_eq_one` turns any one-collar collar geometry with the fixed boundary split into the canonical facewise witness choice, and `exists_planarBoundaryCanonicalWitnessChoice_of_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData` packages the honest-source form where the geometry preserves the source's extracted boundary split. The route file now pushes both positive surfaces to the current theorem-4.9 endpoint: `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_direct` uses the canonical witness choice to feed the split annulus witness-assignment synthesis layer, while `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData` first extracts that canonical witness choice from same-boundary one-collar geometry. The source-facing construction burden may therefore be carried as either the explicit facewise canonical witness choice or the stronger one-collar geometry preserving the source boundary split. The hypothesis is now connected to the canonical Definition 4.8 spanning lane too: `exists_theorem49_kempeGeneratorSubspace_spanning_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_heightOrdered_direct` and `exists_theorem49_kempeGeneratorSubspace_spanning_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_collarLayer_direct` start from canonical witness choice itself, while `exists_theorem49_kempeGeneratorSubspace_spanning_of_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_heightOrdered_direct` and `exists_theorem49_kempeGeneratorSubspace_spanning_of_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_collarLayer_direct` start from same-boundary one-collar geometry. Each wrapper factors through the repaired previous-boundary surface before invoking the height-ordered or collar-layer canonical-generator span bridge, so no separate `Theorem49SubmoduleDualityData` construction remains on these source-side one-collar routes. The corrected boundary-zero target is now available directly as well: `PlanarBoundaryClosedWalkSourceProjection.lean` routes canonical witness choice, the local at-most-one source package, and same-boundary one-collar geometry to `projectedKempeClosureGeneratorSubspace_eq_planarBoundaryZeroSubmodule` and to the corresponding `span_projectedKempeClosureGeneratorFamily_eq_planarBoundaryZeroSubmodule` form. This uses the projected Definition 4.8 generator family, which is the generator source that actually lies inside the selected-boundary-zero submodule, so the source-side statement no longer needs an arbitrary target subspace or raw-generator containment premise. The same file now lifts this endpoint to the route-facing successor-cycle source shell: successor-cycle boundary data plus selected-arc contiguity lowers to the honest closed-walk source package while preserving canonical witness choice, the local at-most-one package, or same-boundary one-collar geometry. The resulting projected-span wrappers include `exists_projectedKempeClosureGeneratorSubspace_eq_planarBoundaryZeroSubmodule_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_canonicalWitnessChoice_heightOrdered_direct` and the corresponding finite-collar and explicit-family variants for canonical witness choice, at-most-one, and one-collar geometry. `PlanarBoundaryClosedWalkSourceRoute.lean` now also has the unprojected raw Definition 4.8 submodule-spanning wrappers on the same successor-cycle source surfaces. The canonical-choice, at-most-one, and same-boundary one-collar successor-cycle packages each feed the height-ordered and finite collar-layer `kempeClosureGeneratorSubspace` spanning bridge, with the target subspace, raw-generator containment, and boundary-zero interpretation still explicit. `PlanarBoundaryClosedWalkSourceRoute.lean` now pushes the same route-facing source shell through the full synthesis package. The successor- cycle canonical-choice and facewise cardinal at-most-one surfaces land directly on `Theorem49BoundaryRootSynthesis`; either surface plus nonempty `selectedBoundaryInteriorEdgeEndpointVertices` lands on `Theorem49BoundaryRootNonemptySynthesis`; and the same-boundary one-collar source geometry now has both direct and nonempty synthesis wrappers. The facewise at-most-one successor-cycle endpoints are `exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_facewiseAtMostOneInteriorEdge_direct` and `exists_theorem49BoundaryRootNonemptySynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_facewiseAtMostOneInteriorEdge_with_nonempty_selectedBoundaryInteriorEdgeEndpointVertices_direct`. These theorems close the packaging gap between the boundary-order source data used upstream and the current theorem-4.9 synthesis API, while leaving the genuine geometric burden exactly at canonical choice, facewise at-most-one, or same-boundary one-collar geometry. The hypothesis is also positively inhabited on the honest diamond-with-triangle source model: `diamondWithTriangleClosedWalkCanonicalWitnessChoice` constructs the canonical facewise witness choice for the boundary split extracted from `diamondWithTriangleClosedWalkAnnulusBoundarySource`, `exists_embedding_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_diamondWithTriangleGraph` packages the corresponding source-level existence, and `diamondWithTriangleGraph_admitsPlanarBoundaryAnnulusWitnessAssignment_via_closedWalkCanonicalWitnessChoice` lowers it to the canonical one-collar witness-assignment endpoint. This model now also has a concrete coloring witness: `diamondWithTriangleTaitEdgeColoring` is proved by `diamondWithTriangleTaitEdgeColoring_isTait` to be a proper nonzero edge coloring, and `diamondWithTriangleEmbedding_theorem49BoundaryRootSynthesis_for_explicitTaitColoring` therefore reaches the theorem-4.9 synthesis surface for an actual named Tait coloring. The same explicit Tait benchmark now also reaches synthesis via the generic at-most-one route: `exists_closedWalkSource_with_atMostOneInteriorEdgePerFace_diamondWithTriangleGraph` packages the local source hypotheses, `diamondWithTriangleClosedWalkAnnulusBoundarySource_` `hasCanonicalWitnessChoice_via_atMostOneInteriorEdgePerFace` reconstructs canonical witness choice from them, and `diamondWithTriangleEmbedding_theorem49BoundaryRootSynthesis_for_explicitTaitColoring_via_atMostOneInteriorEdgePerFace` proves the fixed-embedding endpoint through that local criterion. The new end-to-end package `diamondWithTriangle_closedWalkSource_explicitTait_atMostOne_to_theorem49RawQuotientErrorPackage` then extracts the raw Definition 4.8 representative and selected-boundary-kernel error decomposition for every Kirchhoff chain on that same embedding; `diamondWithTriangle_explicitTait_atMostOne_synthesis_has_empty_selectedBoundaryInteriorCarrier` records that this at-most-one derivation still has empty purified selected-boundary interior carrier on the diamond benchmark;

- the same explicit Tait-positive diamond benchmark is now calibrated against the stronger nonempty-carrier target. Lean packages the whole current positive stack as `diamondWithTriangle_explicitTait_currentSufficientSameEmbeddingGeometry`: the honest source, canonical witness choice, explicit Tait coloring, one-collar geometry, collar-layer/height/well-founded peel data, attached rooted certificate, and current boundary-root synthesis all hold on the same embedding. But `diamondWithTriangle_explicitTait_synthesis_has_empty_selectedBoundaryInteriorCarrier` also proves that the purified selected-boundary interior carrier is empty and that this same explicit coloring does not inhabit `Theorem49BoundaryRootNonemptySynthesis`. So the remaining positive burden is not another current-synthesis witness; it is a Tait-colorable honest source whose purified selected-boundary carrier is nonempty. This is now recorded at the projected endpoint as well: `not_theorem49BoundaryRootNonemptyProjectedSynthesis_diamondWithTriangle` rules out `Theorem49BoundaryRootNonemptyProjectedSynthesis` on the diamond-with-triangle embedding for every coloring, and `diamondWithTriangle_explicitTait_synthesis_blocks_nonemptyProjectedSynthesis` packages the explicit Tait-colored ordinary synthesis endpoint with that projected-endpoint obstruction;

- a separate carrier-calibration benchmark now shows that the Tait and nonempty purified-carrier conditions can be made concrete without using the shared-interior-pair obstruction model as the only positive witness. `Theorem49PlanarBoundaryAnnulusWheelWitnessRegression.lean` defines `wheelWithInnerTriangleGraph`, whose three interior spokes meet a central vertex while a disjoint inner triangle supplies selected-boundary support. The explicit coloring `wheelWithInnerTriangleTaitEdgeColoring` is proved Tait by `wheelWithInnerTriangleTaitEdgeColoring_isTait`, and `selectedBoundaryInteriorEdgeEndpointVertices_nonempty_wheelWithInnerTriangle` proves that the purified selected-boundary interior carrier is nonempty. The bundle `wheelWithInnerTriangle_tait_and_nonempty_selectedBoundaryInteriorCarrier` is intentionally only a Tait-plus-carrier witness, not a positive canonical witness-choice or same-embedding collar-geometry theorem. Lean now proves the negative calibration directly: `exists_two_distinct_interior_edges_on_wheelWithInnerTriangle_boundary` exhibits a wheel face with two distinct interior spokes, `not_nonempty_planarBoundaryCanonicalWitnessChoice_wheelWithInnerTriangle` rules out canonical witness choice for every boundary split on the embedding, and `not_exists_oneCollar_planarBoundaryAnnulusCollarGeometry_wheelWithInnerTriangle` rules out one-collar weak annulus geometry. The bundled theorem `wheelWithInnerTriangle_tait_and_` nonempty_carrier_without_canonical_or_oneCollar therefore records exactly that this model separates the coloring/carrier obligations from the stronger witness-ownership burden;

- the same wheel benchmark has now been calibrated against the face-local at-most-one sufficient condition. Lean proves wheelWithInnerTriangle_face0_interiorEdgeFilter_card: the interior-edge filter on face 0 has cardinality 2. Therefore not_wheelWithInnerTriangle_atMostOneInteriorEdgePerFace blocks the local premise used by PlanarBoundaryCanonicalWitnessChoice.of_atMostOneInteriorEdgePerFace, and not_exists_closedWalkSource_with_atMostOneInteriorEdgePerFace_wheelWithInnerTriangleEmbedding rules out the full fixed-embedding fallback/at-most-one/boundary-rest package for any honest closed-walk source on that embedding. The sharper theorem wheelWithInnerTriangle_tait_and_nonempty_carrier_without_atMostOne_or_canonical_or_oneCollar records the exact status: Tait colorability and nonempty purified carrier hold, while the at-most-one premise, canonical witness choice, and one-collar geometry all fail;

- the wheel obstruction now reaches the raw replacement witness-cover interfaces. Lean proves not_nonempty_planarBoundaryHeightOrderedFacePeelWitnessData_wheelWithInnerTriangle: once a peeled wheel face chooses one central spoke, the other central spoke on that face must be witnessed at strictly larger height; following the three triangular wheel faces forces a strict height cycle through wit01, wit02, and wit03. The same cycle now also blocks the acyclic parent-peeling surface: Theorem49PlanarBoundaryPeeling.lean proves PlanarBoundaryWellFoundedFacePeelWitnessData.toPlanarBoundaryHeightOrderedFacePeelWitnessData using the canonical parent-height rank, and the wheel regression applies this as not_nonempty_planarBoundaryWellFoundedFacePeelWitnessData_wheelWithInnerTriangle. The finite collar-layer interface is blocked by not_nonempty_planarBoundaryCollarLayerFacePeelWitnessData_wheelWithInnerTriangle, because collar-layer data canonically lowers to height-ordered witness-cover data. The same fixed embedding now also blocks weak annulus-collar geometry itself: not_nonempty_planarBoundaryAnnulusCollarGeometry_wheelWithInnerTriangle composes the collar-to-height lowering with the strict height-cycle contradiction, and not_forall_some_acyclicWitnessCover_or_annulusCollarGeometry_of_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_wheelWithInnerTriangle records the corresponding failed universal from Tait colorability plus nonempty purified carrier. This identifies a real construction-side requirement: the positive route must provide a well-founded orientation or selected-boundary break in the interior-face witness dependency relation before even the direct well-founded/height/collar witness surfaces, and hence weak annulus-collar geometry, can be expected;

- the same wheel model now also instantiates the abstract forcing-edge obstruction. In Theorem49ForcingInteriorEdgeObstructionRegression.lean, wheelWithInnerTriangleForcingInteriorEdgeWitness chooses the central spoke wit01; the two incident wheel faces contain selected-boundary support wit12 and wit31. Therefore Lean derives not_nonempty_planarBoundaryAnnulusRootAdjDistancePeelData_wheelWithInnerTriangle_via_forcingInteriorEdgeWitness and not_nonempty_planarBoundaryAnnulusConstructionFaceLayerData_wheelWithInnerTriangle_via_forcingInteriorEdgeWitness. The bundle wheelWithInnerTriangle_tait_nonempty_carrier_and_forcing_obstruction shows that this Tait-colorable, nonempty-carrier benchmark also carries the same obstruction that blocks the current annulus-root and face-layer construction routes on the fixed embedding;

- the cyclic-source at-most-one derivation attempt has also been converted from commentary into a concrete failed universal. The shared-interior-pair honest source now proves `sharedInteriorPair_sipFace0_interiorEdgeFilter_card`: on `sipFace0`, the interior-edge filter consists of the two shared interior edges `sip01` and `sip12`. Therefore `not_sharedInteriorPair_atMostOneInteriorEdgePerFace` blocks the face-local premise, while `route_impossibility_at_most_one_not_derivable_from_cyclic_source` and `not_forall_closedWalkSource_implies_atMostOneInteriorEdgePerFace` in `Theorem49CyclicSourceAtMostOneDerivation.lean` show that even an honest closed-walk annulus source does not imply at-most-one. This records the at-most-one condition as an independent sufficient hypothesis rather than a consequence of cyclic boundary-order data;

- the strengthened-condition derivation note is now theorem-level as well. `Theorem49StrengthenedConditionAtMostOneDerivation.lean` packages the same shared-interior-pair witness with annulus boundary reachability, dart-successor facial geometry, and selected-boundary arcs in `exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_without_atMostOneInteriorEdgePerFace_sharedInteriorPair`. The failed universals `not_forall_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_implies_atMostOneInteriorEdgePerFace_sharedInteriorPair` and `not_forall_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryComponentConstant_implies_atMostOneInteriorEdgePerFace_sharedInteriorPair` show that even adding the boundary-component constancy derived from the honest source fields does not force at-most-one;

- `Theorem49DerivationImpossibilitySummary.lean` now packages the separate route obstructions as `theorem49_route_obstruction_matrix`. The theorem records, in one checked bundle, the cyclic-source failure to imply at-most-one, the chained-diamonds failure of source plus at-most-one to imply nonempty purified carrier, the strengthened successor-cycle failed universal with boundary-component constancy, and the wheel-with-inner-triangle Tait/nonempty-carrier forcing-edge obstruction to the current annulus-root and face-layer routes;

- the positive canonical-witness burden is now inhabited by a second concrete honest source model, not merely by another general constructor. `Theorem49PlanarBoundaryAnnulusHonestGeometryRegression.lean` defines `chainedDiamondsWithTriangleGraph`: two diamond cells chained along one outer boundary component, plus a disjoint triangular inner boundary component. The honest source `chainedDiamondsClosedWalkAnnulusBoundarySource` carries closed facial walks and selected-boundary arcs for all five faces. The direct finite witness `chainedDiamondsClosedWalkCanonicalWitnessChoice` chooses the two interior shared edges `cdt12` and `cdt45` on the two diamond pairs, and one boundary edge on the inner triangle. Lean now proves `chainedDiamondsWithTriangle_interiorEdgeSupport_eq`: the interior edge support is exactly $\{\mathtt{cdt12}, \mathtt{cdt45}\}$, and `chainedDiamondsWithTriangle_atMostOneInteriorEdgePerFace`: each face has at most one interior edge on its boundary. So this model, despite having two distinct interior witness edges across five faces, still satisfies the local face-wise at-most-one condition enabling the canonical-choice constructor. The remaining generic hypotheses are now explicit: `chainedDiamondsCanonicalWitnessEdge_mem` supplies the fallback edge on each face, `chainedDiamonds_nonInteriorOnAmbient` proves that every non-interior face-boundary edge lies on the extracted ambient boundary, and `exists_closedWalkSource_with_atMostOneInteriorEdgePerFace_chainedDiamondsWithTriangleGraph` packages the full local premise. Lean then reconstructs canonical witness choice via `chainedDiamondsClosedWalkAnnulusBoundarySource_hasCanonicalWitnessChoice_via_atMostOneInteriorEdgePerFace` and derives `chainedDiamondsWithTriangleGraph_admitsPlanarBoundaryAnnulusWitnessAssignment_via_atMostOneInteriorEdgePerFace` and `chainedDiamondsWithTriangleGraph_admitsPlanarBoundaryAnnulusCollarGeometry_via_atMostOneInteriorEdgePerFace`. The route-level endpoint theorem now applies to the same package: `chainedDiamondsWithTriangleEmbedding_theorem49BoundaryRootSynthesis_via_atMostOneInteriorEdgePerFace` and `exists_theorem49BoundaryRootSynthesis_chainedDiamondsWithTriangleGraph_via_atMostOneInteriorEdgePerFace` show that the local criterion reaches the current theorem-4.9 synthesis surface, conditional on a Tait coloring, without appealing to the hand-built one-collar object. The same model has now been calibrated against the stronger nonempty-carrier target: `selectedBoundaryInteriorEdgeEndpointVertices_eq_empty_chainedDiamondsWithTriangle` and `not_selectedBoundaryInteriorEdgeEndpointVertices_nonempty_chainedDiamondsWithTriangle` prove that selected-boundary purification deletes all endpoints of `cdt12` and `cdt45`. Combined with `not_exists_taitEdgeColoring_chainedDiamondsWithTriangleGraph`, the status theorem `chainedDiamonds_atMostOne_source_has_empty_carrier_and_no_tait` records that chainedDiamonds contributes at-most-one source geometry, but not the nonempty-carrier or Tait-coloring hypotheses needed by the new endpoint. This obstruction is also recorded as an explicit failed universal in `Theorem49AtMostOneNonemptyCarrierImpossibility`: Lean proves `route_impossibility_at_most_one_does_not_imply_nonempty_carrier`, `not_forall_atMostOneInteriorEdgePerFace_implies_nonempty_carrier`, and `not_forall_closedWalkSource_and_atMostOneInteriorEdgePerFace_implies_nonempty_carrier` from the same chainedDiamonds honest source, so carrier nonemptiness remains an independent hypothesis for the non-vacuous endpoint. The route-facing successor-cycle source shell is now sharper than an independence result. The diamond-with-triangle model carries `diamondWithTriangleDartSuccessorCycleGeometry` with selected-arc contiguity proved by `diamondWithTriangleDartSuccessorCycleGeometry_selectedBoundaryArcOnFace`, satisfies facewise at-most-one, and still has empty purified carrier. Lean packages this as `successor_cycle_at_most_one_and_empty_carrier_independent` and the failed universal `not_forall_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_atMostOneInteriorEdgePerFace_implies_nonempty_carrier`. More importantly, Lean now proves this obstruction at the honest closed-walk level: `selectedBoundaryInteriorEdgeEndpointVertices_eq_empty_of_closedWalkEmbeddingData_and_facewiseAtMostOneInteriorEdge` says that facial closed-walk data plus facewise cardinal at-most-one forces the selected-boundary-purified interior-edge endpoint carrier to be empty on any embedding. Its contradiction forms `not_exists_closedWalkEmbeddingData_and_facewiseAtMostOneInteriorEdge_and_nonempty_selectedBoundaryInteriorEdgeEndpointVertices` and `not_exists_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge_and_nonempty_selectedBoundaryInteriorEdgeEndpointVertices` therefore rule out the current closed-walk at-most-one non-vacuous endpoint outright. Lean also records the constructive contrapositive as `exists_two_distinct_interior_edges_on_faceBoundary_of_closedWalkEmbeddingData_and_nonempty_selectedBoundaryInteriorEdgeEndpointVertices` and the route-facing `exists_two_distinct_interior_edges_on_faceBoundary_of_closedWalkAnnulusBoundarySource_and_hasUnblockedInteriorEndpoint`: any honest closed-walk source carrying the live unblocked endpoint witness must have a face boundary with two distinct interior edges. The route-facing positive annulus-collar package now carries the same necessary condition: `exists_two_distinct_interior_edges_on_faceBoundary_of_closedWalkAnnulusCollarPositiveProjectedGeometryOn` consumes `Theorem49ClosedWalkAnnulusCollarPositiveProjectedGeometryOn`, separating that positive replacement surface from the facewise at-most-one branch before the synthesis endpoint is used. The successor-cycle theorem `selectedBoundaryInteriorEdgeEndpointVertices_eq_empty_of_dartSuccessorCycleEmbeddingData_and_facewiseAtMostOneInteriorEdge` is now the route-facing specialization, not the deepest obstruction. The older direct source existence theorem `exists_embedding_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_chainedDiamondsWithTriangleGraph` and the lowered endpoint `chainedDiamondsWithTriangleGraph_admitsPlanarBoundaryAnnulusWitnessAssignment_via_closedWalkCanonicalWitnessChoice` show that canonical witness choice exists on a larger honest model with two separate interior witness edges; the new at-most-one route shows the generic local criterion itself is inhabited by that same larger model. The same finite witness now builds the named one-collar geometry `chainedDiamondsOneCollarGeometry`; Lean proves `chainedDiamondsOneCollarGeometry_numCollars` and `chainedDiamondsOneCollarGeometry_boundaryData`, packages the source-level one-collar existence theorem `exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_chainedDiamondsWithTriangleGraph`, and records the graph-level admissibility endpoint `chainedDiamondsWithTriangleGraph_admitsPlanarBoundaryAnnulusCollarGeometry_via_closedWalkCanonicalWitnessChoice`. The same concrete source now reaches the downstream certificate branch: `chainedDiamondsOneCollarWitnessOnCurrentBoundary` and `chainedDiamondsOneCollarPreviousBoundaryWitnessGeometry` record the repaired placement surface, and Lean constructs `chainedDiamondsOneCollarCollarLayerFacePeelWitnessData`, `chainedDiamondsOneCollarHeightOrderedFacePeelWitnessData`, `chainedDiamondsOneCollarWellFoundedFacePeelWitnessData`, and `chainedDiamondsOneCollarAttachedRootedFacePeelCertificate`. The graph-level endpoints `chainedDiamondsWithTriangleGraph_admitsPlanarBoundaryAttachedRootedFacePeelCertificate_via_oneCollarGeometry` and `exists_theorem49BoundaryRootSynthesis_chainedDiamondsWithTriangleGraph_via_oneCollarGeometry` show that this larger honest-source witness already lands on the current certificate and theorem-4.9 boundary-root synthesis surfaces. The coloring side is explicitly calibrated by `not_exists_taitEdgeColoring_chainedDiamondsWithTriangleGraph`: the shared vertex 3 has four incident edges, so a Tait coloring would require four pairwise distinct nonzero colors although the color space supplies only three. Hence the chained model is a larger positive source/collar/certificate witness, while its synthesis theorem is vacuous as a Tait-coloring instance;

- that blocker now extends to the whole one-collar weak-geometry branch. The new theorem `not_nonempty_planarBoundaryAnnulusWitnessAssignment_of_numCollars_eq_one_and_exists_two_distinct_interior_edges_on_faceBoundary` in `Theorem49PlanarBoundaryAnnulusWitness.lean` shows that any one-collar witness assignment already fails when one face carries two distinct interior edges. Then `Theorem49PlanarBoundaryAnnulusGeometry.lean` lifts this to `not_exists_planarBoundaryAnnulusCollarGeometry_of_numCollars_eq_one_and_exists_two_distinct_interior_edges_on_faceBoundary`, and `Theorem49PlanarBoundaryAnnulusHonestWitnessRegression.lean` instantiates it on the same honest source model as `not_exists_oneCollar_planarBoundaryAnnulusCollarGeometry_sharedInteriorPair`. So the live weak-collar route from honest source data is now blocked on the entire one-collar branch, not only on the canonical decomposition recovered from the source;

- the same witness first blocked the fixed-source-split two- and three-collar repairs through the current-boundary bridge `not_exists_planarBoundaryAnnulusCollarGeometry_of_not_exists_planarBoundaryAnnulusCurrentBoundaryData_of_numCollars_eq_and_boundaryData`. That was already enough to show that the source boundary split is not a free weak-collar geometry;
- the stronger regression no longer fixes the boundary split. The generic terminal-layer theorem `false_of_mem_terminal_collarFaces_and_exists_two_distinct_interior_edges_on_faceBoundary` shows that a terminal collar face cannot carry two distinct interior edges. On `sharedInteriorPairAnnulusEmbedding`, both outer faces carry the two interior edges `sip01` and `sip12`, so `terminal_collarFaces_eq_singleton_sipFace2` forces every terminal collar to be exactly the inner triangular face. Then `false_of_intermediateBoundary_before_singleton_sipFace2` rules out any predecessor stage whose remainder is this singleton, yielding `not_exists_twoCollar_planarBoundaryAnnulusCollarGeometry_sharedInteriorPair` and `not_exists_threeCollar_planarBoundaryAnnulusCollarGeometry_sharedInteriorPair` without requiring the geometry's boundary split to equal `sharedInteriorPairBoundaryData`;
- the remaining "start at four collars" escape hatch is still closed by the finite-face bound `PlanarBoundaryAnnulusCollarGeometry.numCollars_le_card_faces`. The honest source file specializes it as `numCollars_le_three_of_planarBoundaryAnnulusCollarGeometry_sharedInteriorPair`, since `sharedInteriorPairAnnulusEmbedding` has exactly three ambient faces. Combining that bound with the one-, two-, and three-collar blockers gives `not_nonempty_planarBoundaryAnnulusCollarGeometry_sharedInteriorPair`. Thus `closedWalkAnnulusBoundarySource_without_annulusCollarGeometry_sharedInteriorPair` and `exists_embedding_closedWalkAnnulusBoundarySource_without_annulusCollarGeometry_sharedInteriorPair` now witness an honest closed-walk annulus source with no weak annulus collar geometry on the same embedding at all, regardless of how a later repair tries to choose the annulus boundary split. The failed universal same-embedding converse is `not_forall_exists_planarBoundaryAnnulusCollarGeometry_of_closedWalkAnnulusBoundarySource_sharedInteriorPair`. The same absence of weak collar geometry now directly refutes the source-alone previous-boundary route: `not_nonempty_planarBoundaryAnnulusPreviousBoundaryWitnessGeometry_sharedInteriorPair` projects a hypothetical repaired previous-boundary witness back to collar geometry, while `closedWalkAnnulusBoundarySource_without_previousBoundaryWitness_sharedInteriorPair` and `not_forall_`

exists_planarBoundaryAnnulusPreviousBoundaryWitnessGeometry_of_closedWalkAnnulusBoundarySource_sharedInteriorPair package the concrete failed implication. This does not settle the stronger question with weak collar geometry already present; it records that honest closed-walk source data alone is insufficient. The corresponding repaired source-plus-collar investigation now has no placeholder theorem: Lean proves honest_source_plus_collar_geometry_extending_previous_boundary_witness_iff_witnessOnCurrentBoundary, which identifies the exact remaining fixed-collar obligation as WitnessOnCurrentBoundary. That obligation is now known not to follow from an arbitrary weak collar on an honest source embedding. The connected chained-diamonds source admits the deliberately mismatched two-collar geometry chainedDiamondsBadTwoCollarGeometry: the terminal triangular face chooses cdt97 while its previous intermediate boundary is the singleton cdt45. Lean proves not_witnessOnCurrentBoundary_chainedDiamondsBadTwoCollarGeometry, packages the source-plus-bad-collar witness as exists_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_without_witnessOnCurrentBoundary_chainedDiamonds, and refutes the universal implication as not_forall_honest_source_plus_collar_geometry_forces_previous_boundary_witness. It also proves not_nonempty_previousBoundaryWitnessGeometry_extending_chainedDiamondsBadTwoCollarGeometry, so this exact weak collar cannot be repaired without changing the collar geometry. Thus the repaired lane must use a collar geometry tied to the source boundary split, carry WitnessOnCurrentBoundary explicitly, or derive the collar from stronger annular geometry. The same module now proves the sharper source-side calibration exists_atMostOne_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_without_witnessOnCurrentBoundary_chainedDiamonds, not_forall_atMostOne_source_plus_collar_geometry_forces_previous_boundary_witness, and not_forall_canonicalWitnessChoice_source_plus_collar_geometry_forces_previous_boundary_witness: the local at-most-one criterion and source-level canonical witness choice are not enough when the weak collar is supplied independently rather than built from that source data;

- the obstruction now reaches the weaker witness-cover surfaces that no longer require annulus collar geometry. In the same regression file, not_nonempty_planarBoundaryHeightOrderedFacePeelWitnessData_sharedInteriorPair shows that the shared interior pair creates a strict height cycle: a peeled face choosing sip01 needs a strictly later peeled face choosing sip12, and that face needs a strictly later peeled face choosing sip01, but only sipFace0 and sipFace1 can choose either shared edge. The finite collar-layer witness-cover form is blocked by not_nonempty_planarBoundaryCollarLayerFacePeelWitnessData_sharedInteriorPair, because it canonically lowers to the height-ordered surface. The same prefix cycle has now been abstracted as not_exists_mem_of_twoWitnessPrefixCycle, which allows repeated faces in the peel list and still rules out a finite mutual dependency. The support-level form not_mem_facePeelWitnessSupport_pair_of_twoWitnessPrefixCycle now records that both mutually dependent witness edges are absent from the resulting witness support. Instantiating it gives not_nonempty_attachedBoundaryRootedFacePeelCertificate_sharedInteriorPair for the raw attached-face BoundaryRootedFacePeelCertificate, and not_nonempty_planarBoundaryWellFoundedFacePeelWitnessData_sharedInteriorPair follows because well-founded-parent witness data lowers to that certificate. Thus closedWalkAnnulusBoundarySource_without_heightOrderedFacePeelWitnessData_sharedInteriorPair and `closedWalkAnnulusBoundarySource_without_collarLayerFacePeelWitnessData_sharedInteriorPair` now sit beside `closedWalkAnnulusBoundarySource_without_attachedBoundaryRootedFacePeelCertificate_sharedInteriorPair` and `closedWalkAnnulusBoundarySource_without_wellFoundedFacePeelWitnessData_sharedInteriorPair`, recording that honest closed-walk annulus source data alone does not force even the raw same-embedding attached certificate accepted by the current Theorem 4.9 synthesis package. The same obstruction model is now Tait-colorable: `sharedInteriorPairTaitEdgeColoring` is proved by `sharedInteriorPairTaitEdgeColoring_isTait`, and `closedWalkAnnulusBoundarySource_and_taitEdgeColoring_without_attachedBoundaryRootedFacePeelCertificate_sharedInteriorPair` packages the honest source, an actual Tait coloring, and failure of the same-embedding attached certificate together;

- this attached-certificate obstruction now also hits the route-facing successor-dart source shell. The same finite model defines `sharedInteriorPairDartSuccessorCycleGeometry` and proves the induced selected-boundary arc condition `sharedInteriorPairDartSuccessorCycleGeometry_selectedBoundaryArcOnFace`. The witness theorem `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_without_attachedBoundaryRootedFacePeelCertificate_sharedInteriorPair` packages boundary reachability, successor-cycle facial data, selected-boundary arc contiguity, and failure of the raw attached certificate on one embedding. The failed converse `not_forall_exists_attachedBoundaryRootedFacePeelCertificate_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_sharedInteriorPair` now has a Tait-calibrated companion: `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_without_attachedBoundaryRootedFacePeelCertificate_sharedInteriorPair` and `not_forall_exists_attachedBoundaryRootedFacePeelCertificate_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_sharedInteriorPair` show that even adding an actual Tait edge coloring to the route-facing successor-dart source shell does not force the same-embedding attached certificate. The bundled theorem `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_without_currentSufficientSameEmbeddingGeometry_sharedInteriorPair` packages simultaneous failure of height-ordered witness data, collar-layer witness data, the raw attached certificate, well-founded witness data, and annulus collar geometry on that same Tait-colorable source. The companion `not_forall_exists_some_currentSufficientSameEmbeddingGeometry_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_sharedInteriorPair` records that the successor-cycle source shell plus Tait coloring is still not enough to force even one of the currently sufficient same-embedding geometry endpoints. This obstruction is now non-vacuous on the purified carrier: `vertex_one_mem_selectedBoundaryInteriorEdgeEndpointVertices_sharedInteriorPair` and `selectedBoundaryInteriorEdgeEndpointVertices_nonempty_sharedInteriorPair` show that vertex 1 survives selected-boundary purification, while `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_without_currentSufficientSameEmbeddingGeometry_sharedInteriorPair` and `not_forall_exists_some_currentSufficientSameEmbeddingGeometry_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_sharedInteriorPair` show that adding this nonempty carrier hypothesis still does not force any of the current sufficient same-embedding geometry endpoints. The same source package now has matching failed converses for `PlanarBoundaryAnnulusCollarGeometry`, `PlanarBoundaryHeightOrderedFacePeelWitnessData`, `PlanarBoundaryCollarLayerFacePeelWitnessDat` and `PlanarBoundaryWellFoundedFacePeelWitnessData`, so this stronger route-facing source shell does not force any of the currently sufficient same-embedding geometric endpoints on the shared-interior-pair witness. `Theorem49PositiveGeometricSourceReplacementRegression.lean` now reflects that obstruction at the newer positive package layer as well: the same witness refutes both `Theorem49ClosedWalkAnnulusCollarPositiveProjectedGeometryOn` and `Theorem49SuccessorCycleAnnulusCollarPositiveProjectedGeometryOn`. The honest closed-walk and successor-source failed converses are now factored through reusable generic source principles in `PlanarBoundaryClosedWalkSource.lean` and target-specific wrappers in `Theorem49PlanarBoundaryAnnulusGeometry.lean`, with the shared-interior-pair model supplying the concrete missing-target witnesses;

- the honest long-term frontier is still to derive this ordered or cyclic data from real embedding-side boundary walks rather than postulating it.

So the present formalization is no longer hiding the gap. The gap is explicit, localized, and regression-tested.

4 Relevant Lean files

- `Mettapedia/GraphTheory/OrderedPlanarEmbeddingWithBoundary.lean`
- `Mettapedia/GraphTheory/OrderedPlanarEmbeddingWithBoundaryRegression.lean`
- `Mettapedia/GraphTheory/WalkPlanarEmbeddingWithBoundary.lean`
- `Mettapedia/GraphTheory/WalkPlanarEmbeddingWithBoundaryRegression.lean`
- `Mettapedia/GraphTheory/FourColor/PlanarBoundaryOrderedEmbedding.lean`
- `Mettapedia/GraphTheory/FourColor/PlanarBoundaryFaceBoundaryRunRegression.lean`
- `Mettapedia/GraphTheory/FourColor/Theorem49ClosedWalkOuterRootObstruction.lean`